One Frozen DINOv3, Wide LoRA and a Tiny-Object Cascade: Multi-Task Learning on SAR Images under a 40 M Budget

CS701 Team 7 (`cs701-team07`)
Junkai Chen, [other authors: TBD]
Singapore Management University

Abstract

One model must classify a SAR image into nine scene classes and detect its objects, half of which are smaller than a 16×16 ViT patch, under limits of 130 M total and 40 M trainable parameters. Our model keeps a DINOv3 ViT-B/16 frozen and spends the budget on LoRA rank 80 and a three-stage Cascade R-CNN built for tiny boxes. It reached $\Delta m = 56.49$ on the Phase 1 validation board (macro-F1 95.27, mAP 55.10). The Phase 2 entry uses the same weights plus two post-processing additions validated locally, and scored $\Delta m = 61.25$ on the test set (macro-F1 96.96, mAP 55.74, AP50 75.19), third of eight on the board on 5 November, before it locks. Almost all the gain over the reference comes from detection. We report what each part adds: leave-one-out ablations on a held-out split over several seeds, which we ran specifically for this report, and a post-processing ablation on full-length runs (seed 0 finished; further seeds TBD). We also report what failed, including results that won on a local split and vanished on the board.

1. Introduction

Each image has one class label and one or more boxes of that class. Teams are ranked by $\Delta m = 50\,[(F_1 - F_1^{\text{ref}})/F_1^{\text{ref}} + (\text{mAP} - \text{mAP}^{\text{ref}})/\text{mAP}^{\text{ref}}]$, with reference scores 93.31 / 26.13 on val and 93.99 / 25.41 on test. On val, one mAP point is worth 1.91 Δm and one F1 point 0.54 Δm. A point of mAP therefore pays 3.6 times as much as a point of F1, so we spent nearly all our effort on detection.

Three properties of the data shaped the design. (1) Boxes are tiny: at the 512 px input, 18.4 % have side <8 px and 50.5 % have side <16 px. (2) Localisation is the bottleneck, not recall alone: our final local AP falls from about 75 at IoU 0.5 to 22 at IoU 0.9. (3) Annotations carry structure a model can exploit: a sub-pixel offset and, for some class–size cells, an integer coordinate lattice (§4.5).

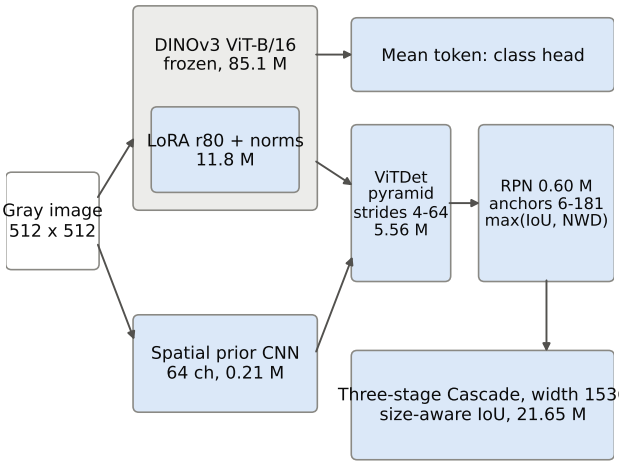


Figure 1. The final model. Only the blue blocks train (39.87 M). Each image passes through once at 512×512.

2. Related work

LoRA [5] and MTLoRA [1] adapt frozen weights at low cost; we use a single wide LoRA. For tiny objects, NWD [10] replaces IoU in assignment, Cascade R-CNN [3] refines boxes, ViTDet [7] builds a pyramid from one ViT map, and ViT-Adapter [4] adds a CNN spatial prior. Soft-NMS [2] and weight averaging along a trajectory [6] are the two ideas behind our largest zero-cost gains. Δm is the usual multi-task summary [8, 9].

3. Method

Backbone. Timm `vit_base_patch16_dinov3.lvd1689m`, frozen; the RGB patch kernel is summed to one gray channel. LoRA rank 80, α=160 wraps `qkv`, `proj`, `fc1` and `fc2` of all 12 blocks, and LayerNorms train.

Neck. A ViTDet pyramid gives strides 4–64 at 256 chan-

nels. A 64-channel randomly initialised CNN (210,688 parameters) reads the raw image and adds to strides 4 and 8 through zero-initialised 1×1 convolutions, restoring pixel detail a stride-16 token map lacks.

RPN. Anchors of 6–181 px at cell centres, matched by $\max(\text{IoU}, \text{NWD})$ with $C{=}8$, because a 2 px offset already sends the IoU of an 8 px box close to zero.

Cascade. Three class-agnostic stages of width 1536 with aligned RoIAlign. Fixed IoU thresholds of 0.5/0.6/0.7 leave tiny objects without positives at later stages, so thresholds depend on box side: .40/.45/.50 below 8 px, .45/.50/.55 for 8–16 px, .50/.55/.65 for 16–32 px.

Flip bias. Boxes carry a systematic offset of about -0.3 px. A plain horizontal flip mirrors it, so targets swing by twice the bias; `-flip-bias -0.3` moves flipped boxes back.

Training. Flips only; AdamW with head LR 10^{-4}, backbone LR $3{\times}10^{-4}$, weight decay 0.05; one warm-up epoch, then a 200-epoch cosine; global batch 16; all 9,392 labelled images for the submitted model.

Checkpoint averaging. The nine checkpoints at epochs 100, 110, …, 180 of one run are averaged tensor by tensor into one set of weights (234 tensors, FP64 accumulation). The result is one model with one forward pass, not a prediction ensemble. LoRA makes the average inexact, since $\overline{BA} \neq \bar{B}\bar{A}$; the measured gap between e180 and e200 was 0.002 %.

Post-processing uses only that one forward pass. Boxes are the mean of the three stages, shifted by -0.2 px. With p the stage-averaged box probabilities and p_{img} the image classifier, box scores use class-agnostic fusion $s_c = (1 - p_{\text{bg}})\, p_{\text{img}}(c)$: objectness is dropped, and each image's class prior removes false positives of other classes without discarding true boxes in misclassified images. Then per-class Gaussian soft-NMS with $\sigma{=}0.5$, RPN top 2000, threshold 10^{-5}, 100 boxes per image. The image label is $\arg\max_c[\log p_{\text{img}}(c) + 2\log\max_i p_i(c)]$.

Final entry additions (never scored on val). (a) `min_keep` 100 keeps the best 100 boxes of an image when too few clear the threshold. (b) Lattice snapping: predicted corners of aircraft, car and all 256 px images are rounded to integers in original pixels. For wind turbines, a per-image fractional offset is estimated as a circular mean over confident boxes and applied only when the concentration $R \geq 0.5$. Scores and order do not change.

4. Experiments

4.1. Validation and how we chose the entry

Split. `prepare_course_data.py` splits the train set by class with seed 0 into 7,512 / 940 / 940, used for screening up to 11-01. The ablations here use a second split that shares a pool with it: `vallike` (992 images from the old holdouts, sampled to match the (predicted class, image size) mix of the official val set) and `trainvl` (the other 8,400). Official val and test images were only ever predicted; their predicted class counts set the sampling mix.

Choosing the Phase 2 entry. We took the Phase 1 recipe with the highest official val Δm, then added two steps each found positive on local holdouts by paired bootstrap, combined $+0.30$ mAP [+0.20, +0.41]. A twin trained on `trainvl` confirmed that the averaging window sat within noise (0.1 mAP) of the best alternatives. These selections reused the same data many times, and many Phase 1 choices were made on the val board itself, so our val score is optimistic.

Noise. One run of the 36-epoch C36 configuration varies by ±0.70 local Δm (37 seeds); sampling alone gives an SD of about 2 Δm between test and val. Post-processing comparisons are paired on the same weights and images, so differences of 0.1–0.3 Δm can be real there, while differences between runs below about 1 Δm are not.

4.2. Official progression

Table 1 lists the board milestones. Most steps change several things, so attribution comes from §4.3. Between F2 and SC2, wider adaptation and head account for the largest step; from there, a slower cosine helped twice, and the last 3.1 Δm cost no training at all. We finished Phase 1 third of nine, behind Team 9 (67.16) and Team 6 (65.85). Team 6's AP50 was only 0.8 above ours while its mAP was 4.9 above, so its lead lies almost entirely at strict IoU. On test, our entry scored 61.25 against Team 6's 71.57 and Team 9's 71.40; the gap stays almost entirely in mAP (55.74 against 61.21 and 61.08), while our macro-F1 of 96.96 is above both (96.11, 96.27). Test and val use different references and images, so 61.25 and 56.49 are not directly comparable.

4.3. What each part adds

Model parts. Table 2 removes one part at a time from the final recipe: 36-epoch cosine on `trainvl`, scored on `vallike` with the full FINAL post-processing on the last checkpoint. These runs were trained for this report. Short runs understate parts that pay late, as the slower-schedule gains below show, so they rank parts rather than predict the full-length gain.

Earlier ladder. On the 7,512 split, raw local Δm against our own reference run rose: DINOv3 instead of ImageNet-21k $+8.1$; LoRA r32 on all linear layers plus norms $+18.1$; Cascade $+21.0$; NWD with small anchors $+19.7$ (both 24 epochs); both at 36 epochs $+31.3$; at 48 epochs $+38.6$. Against the C36 mean of 37 seeds with final post-

Date	Entry	Δm	F1	mAP	Change
10-20	V2, 120-ep cosine, e100	38.23	95.06	45.62	first long run, 90 % data
10-21	F1, all data, 100 ep	41.26	95.90	46.97	all data, better post-proc.
10-27	F2	44.84	95.46	48.97	size-aware IoU, flip bias, shift
10-28	SC2, e130 of 150	51.86	95.35	52.66	LoRA r80, head 1536, longer
10-31	BP7, e130 of 160	52.78	95.35	53.15	spatial prior, slower cosine
11-01	cos200, e160	53.41	95.02	53.56	slower cosine again
11-01	e180, σ.5, obj^0, RPN 2k, λ2	54.83	95.09	54.29	post-processing only
11-01	avg e100/120/140/160	55.41	95.20	54.56	checkpoint averaging
11-01	avg e100–180 (9), hard	56.41	95.27	55.06	wider window + hard fusion
11-01	**same, class-agnostic**	**56.49**	**95.27**	**55.10**	**posted, Phase 1**
11-03	**test: + min_keep 100, snap**	**61.25**	**96.96**	**55.74**	**posted, Phase 2** (test reference)

Table 1. Official scores (Phase 1 val, reference 93.31 / 26.13; the last row is test, reference 93.99 / 25.41; seed 0, 1 run each). V2 and F1 were scored on Codabench, the rest on the course leaderboard.

Variant	Train. M	n	mAP	F1	$\Delta\Delta m$
Final recipe	39.87	1	45.67	96.71	—
no spatial prior	39.66	1	45.85	95.62	-0.25
no size-aware IoU	39.87	TBD			
no flip bias (shift -0.4)	39.87	TBD			
no NWD matching	39.87	TBD			
no Cascade (Faster R-CNN)	32.16	TBD			
reference anchors	39.86	TBD			
LayerNorms frozen	39.83	TBD			
LoRA r32, head 1024	24.02	TBD			

Table 2. Leave-one-out on `vallike` (36-epoch cosine, FINAL post-processing, last checkpoint). $\Delta\Delta m$ is variant minus final recipe, paired by seed; mean$\pm$std where $n > 1$.

processing, size-aware IoU added $+1.69$ (z=2.4; wind turbine AP $+6.5$; AP below 8 px from 8.2 to 12.9), flip bias $+1.23$ (z=1.75), and both $+2.26$ (z=3.2). LoRA r80 with head 1536 added $+5.92$ on top. These screens are single-seed.

Schedule. On official val, cosine length at a fixed stopping epoch gave $+1.29$ (130→160, 7,512 images, e130) and $+1.02$ on the full data, the second also changing the data loader. Then 160→200 gave $+0.63$ at e160 against e130 of cos160. Halving the LR lost 7.2 local Δm; e200 lost 0.12 to e180.

Post-processing and averaging. Table 3 is a cumulative ablation on the full-length runs (seed 0 finished; seeds 1 and 2 TBD). It reproduces the order of the official steps, with one disagreement: the official gain for objectness power 0

Step (cumulative)	mAP	F1	ΔmAP	$\Delta\Delta m$
e180, σ.75, obj$^{\cdot 25}$, RPN 1k, λ1	57.00	96.46	—	—
σ 0.5	57.15	96.46	+0.15	+0.29
objectness power 0	57.56	96.46	+0.41	+0.79
RPN top 2000	57.60	96.46	+0.04	+0.07
λ 2	57.60	96.65	+0.00	+0.10
average e100–180	58.37	96.53	+0.78	+1.42
hard fusion	58.53	96.53	+0.16	+0.31
class-agnostic (vs hard)	58.59	96.53	+0.06	+0.11
min_keep 100 (vs class-agn.)	58.61	96.53	+0.02	+0.03
snap = FINAL	58.81	96.53	+0.20	+0.39

Table 3. Cumulative post-processing and averaging on full-length runs (n=1 seed; `trainvl` → `vallike`). Step gains are paired on one seed's weights; Δm uses the val reference, on a local split, so its absolute level is not comparable to the board.

was larger than the local one. Class-agnostic fusion beats hard filtering because hard filtering discards every true box in a misclassified image.

4.4. What did not work

Table 4 lists the main failures, single seed unless stated. The context result is the most instructive: $+2.26$ local Δm on one holdout, -0.12 on the other and -0.03 on the board, with harbor AP swinging from $+3.7$ to -1.3 between the two. It is why we stopped trusting one holdout. Several negative verdicts are also weaker than they read. The LR $\times 1.25$ run was stopped at epoch 69, and E09b's fall was never attributed to one of its several changes.

4.5. Findings in the data

Annotation lattice. All training boxes of aircraft and car, and of every 256 px image, have integer x, y, w, h in original pixels. About 40 % of wind-turbine images put all boxes on one lattice with a shared fractional offset. Snapping predictions is the only data-mining finding that survived testing: it adds $+0.20$ and $+0.36$ mAP on the two holdouts (paired bootstrap $P \approx 1$). We assume test shares this lattice. The Phase 2 score does not isolate the effect, because only the combined entry was scored.

Near-duplicates. By frozen DINOv3 cosine, about 10 % of holdout images have a near-copy in train. Local mAP is 83 on these, against 40 for images with no close neighbour (similarity <0.94), and playground AP is 87 versus 12. This explains part of the local–official gap. Transferring a neighbour's boxes in a recall-only mode gained $+0.52$ / $+0.29$ mAP locally, but it needs training images at test time and a second network, so we did not use it.

Weak classes. Bridge stays at AP 8 and playground at 33 on `vallike` even after full training. On the earlier SC1 model, only 49 % of bridge boxes had a top-1000 proposal at IoU 0.5. Size alone does not explain it, since wind tur-

Idea	Result	Note
E09b wide LoRA + regularisers (24 ep)	raw local Δm 6.18 vs 18.08	causes never separated
ConvNeXt-T full fine-tune (24 ep)	6.78; $AP_{<8}$ 1.9	below ViT + LoRA
Prior + multi-level neck (24 ep)	5.14	prior alone later won
D4 + copy-paste; long D4, zoom	7.13; behind plain at all epochs	augmentation hurt
Overlapping patches, stride 8	-0.99, $2.6\times$ compute	oil tank -6.8, harbor -4.6
EMA 0.9998; IoU branch (36 ep)	-0.13; -0.48	lead vanished as LR$\to$0
RoI batch 1024	-4.02	cause not studied
Bridge/playground: class weights, oversampling, partial-neg	bridge $\leq +0.9$ AP; oversampling -4.15	wind -10 AP
Classification-only RoI context (160 ep)	$+2.26$, -0.12, board -0.03	did not transfer
LR $\times 0.5$; head/backbone hybrid	-7.21; -2.77	early lead lost by e100
LingBot-B backbone	board 39.23 at e60 (BP7 e80: 48.40)	paused, no fair pair
Box voting; no 100-box cap	val mAP -0.03 to 0	no effect
Rank, size priors; per-class bias tables; refinement	-3.1 to $+0.06$ mAP	noise or harm
kNN label transfer (local only)	$+0.52/+0.29$ mAP	test-time data, rules; not used

Table 4. Tried and dropped. Numbers are local Δm unless marked; screens are single-seed, so read small values with the ± 0.70 noise of §4.1 in mind.

bines are smaller and reach 79. Class weights, oversampling and partial-negative losses moved bridge by at most $+0.9$ and cost other classes more. Whether unlabelled same-class objects limit these classes is an unverified hypothesis from viewing 12 images.

5. Efficiency and required disclosures

Table 5 gives the items the Terms ask for. We trade parameters for score: 39.87 M of the 40 M limit. The cheaper 23.8 M F2 scored 44.84 at 3.6 h on two GPUs, so about 7 Δm of the path to 51.86 came with 16 M more trainable parameters and more epochs together. Every zero-cost step after cos200 needed no extra training time.

6. Limitations and conclusion

Averaging, fusion, snapping and `min_keep` gains are measured with seeds on one held-out split that also shaped earlier decisions. The 36-epoch ablations do not measure full-length effects, and neither GIoU with an IoU branch (stopped at epoch 145) nor photometric augmentation was evaluated at full length. We also measured no fix for the strict-IoU gap of about 5 mAP to the leading teams.

A strong generic backbone, frozen and adapted with wide LoRA, plus a detector rebuilt for sub-patch objects carried most of the gain; patient schedules and careful use of one forward pass supplied the rest. The most useful habit was to count a local gain only after a second holdout or the board agreed.

Use of AI tools. Most code and experiment execution were done with AI coding assistants (Claude Code and OpenAI Codex), directed by the team. The complete experiment log is in the appendix.

Item	Value
Parameters	**125,076,960 total, 39,867,360 trainable**: frozen ViT 85.14 M; LoRA 11.80 M; norms 0.04 M; neck 5.56 M + prior 0.21 M; RPN 0.60 M; Cascade 21.65 M; class head 0.01 M
GPU	2 $\times$ NVIDIA RTX 6000 Ada (46,068 MiB), shared with other users
Final training	12.97 h to e180 on 2 GPUs (25.9 GPU-h); the run continued to e200 at 14.14 h. Averaging: 30 s CPU
Project GPU-h	About TBD on this server, tuning included; RTX 3080 runs of teammates not counted
Peak GPU memory	Train 12.07 GiB per GPU (gradient checkpointing, 8 per GPU); inference 1.15 GiB allocated
Inference	21–25 ms per image at batch 16 (bf16, including post-processing, GPU shared); 60 ms at batch 2 under load on the test run
Input	One gray image, stretched to 512×512; train mean/std 0.2526/0.2133; one pass, no tiling, no TTA
Pretrained	**Final:** DINOv3 ViT-B/16 LVD-1689M (timm). **Exploration only:** ImageNet-21k ViT-B/16; DINOv3 ConvNeXt-T; DINOv3 ViT-S/16 with YOLO; TerraMind; LingBot-Vision-B/16. None trained on SAR
Code	Base: TA reference repo; torchvision, timm, pycocotools. Ours: LoRA extensions, Cascade with size-aware IoU, NWD matching, spatial prior, post-processing, averaging, snapping, evaluator
Validation	§4.1. Official val and test only predicted
Seeds	Submitted model: 1 run. Final recipe, full length: 1 finished, 2 running (TBD) runs. Ablations: Table 2. C36: 37 runs, ± 0.70

Table 5. Items required by the Terms.

References

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Appendix: Complete Experiment Log

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A. How to read this

This appendix holds what the main report summarises: every idea tried, why, how it was implemented, what it gained on which scale, and why it was kept or dropped. It is a factual record. It makes no recommendations. Line-level sources for every number are in `experiments/report_final_20261105/ledgers/ledger_{A_1013_1027,B_1027_1031,C_1101_1105}.md`.

A.1. Evidence

Unless marked otherwise, a number was read from a run artifact (`metrics.json`, `metrics_pp.json`, `log.txt`, `events.log`, a zip, or code) or recomputed on CPU from saved raw dumps. Weaker evidence is marked as follows.

- † **Recorded only in session notes**: a statement in a working session, with no underlying artifact found.
- ‡ **Repository documents only**: stated in a project document, with no run folder on this server. All Rayner-line numbers are of this kind.
- Numbers described as *session tool output* were printed by a tool during a working session and are preserved in its log, but not as a separate file.
- Official scores were entered by the team from board screenshots or pasted JSON and then recorded in files such as `official_scores_phase1.json`. Screenshots hold two decimals. The assistants never uploaded anything; all uploads were manual.
- *Derived* means simple arithmetic on cited numbers.

Paths use these abbreviations. Times are SGT (UTC+8) unless marked.

Table 6. Path abbreviations.

Short	Path
`$RUNS`	`/mnt/hdd/junkaichen/CS701/cs701_runs`
`$EXP`	`experiments/` in the repository
`$RPT`	`experiments/report_final_20261105/` (copies of the bootstrap evidence in `prior_evidence/`)
`$BK44`	`backups/44.84-DINOv3-LoRA-Cascade-NWD-SizeIoU-FlipBias-shift0.2/`
`$BK28`, `$BK51`	`backups/` folders whose names start with 28.90 and 51.86
`$BK56`	`backups/56.49-full-cos200-avgE100-180-DINOv3-LoRA80-Head1536-SP64-Cascade-NWD-SizeIoU-FlipBias-agn-soft0.5-obj0-shift0.2/`
`$PH2`	`.runtime/phase2/`
`$SUB`	`submissions/`

A.2. Metric scales. Never compare across these

Table 7. Metric scales used in this log.

Scale	Definition	Example
Official val Δm (Phase 1)	$50[(F_1 - 93.31)/93.31 + (\text{mAP} - 26.13)/26.13]$ on 1,426 official val images. 1 mAP point = 1.9135 Δm, 1 F1 point = 0.5358 Δm, ratio 3.57	Posted entry 56.49
Official test Δm (Phase 2)	Same form with references 93.99 / 25.41 on 2,065 images. 1 mAP point = 1.9677 Δm, 1 F1 point = 0.5320 Δm	No Team 7 score is recorded in project files (§E.13)
Local Δm, scale **L**	Reference = our single run E00 (val 91.50 / 22.41, test 91.61 / 23.31). Mean of local val and local test (940 images each). Post-processing version stated	C36 v3: L 35.46
Scale **LT**	L, local test only, for runs trained on train+val	V2 e70: LT 52.9
Scale **BP**	Same reference as L. Mean of local val and test, shift 0, post-processing v3′. Same construction as L; AA rounds to 37.82 in one source and 37.83 in another	AA 37.83, SC1 43.75
Scale **LR**	Same E00 reference. Local val only (test in a separate file), shift −0.2, submission post-processing	lr1x 68.70
Scale **Ry**	Rayner's documents: official val references applied to local-split scores. On this scale E00's raw test score would be −6.31 and E04r's raw test +10.11	Plain YOLO26-s +1.4
Local holdout mAP	COCO mAP on local val, local test (940 each) or merged (1,880); model W3 (§E.5), trained on 7,512	agn: 52.907 / 53.780
vallike mAP	COCO mAP on the 992-image `vallike` holdout; twins trained on 8,400 (§E.11)	TWIN1 FINAL: 58.81

(continued)

Scale	Definition	Example
vallike Δm	Official val references applied to vallike scores (Tables 2, 3). Paired differences are usable; the absolute level is not comparable to the board	TWIN1 FINAL: 64.26
Proxy score	$1.9135\,\text{mAP} + 0.5358\,F_1$ with no reference subtracted. Not a Δm	Used only in `proxy_eval.py`

Three examples of why these must not be mixed. (1) lr1x e130 is 68.70 on scale LR and 45.83 on the board; that gap is mainly a difference of references, not a loss. (2) V2 e100 has local test F1 95.59 and mAP 48.91, which gives +44.81 under official references, against 38.23 on the board. (3) The same W3 weights score about 2.4 mAP higher on vallike than on merged local holdouts (§E.11). The reference values are rounded as well: applying the formula to the exact scores of the posted entry gives 56.4911, while the board shows 56.486765.

A.3. Post-processing versions

Table 8. Post-processing versions. Thr = score threshold; σ = soft-NMS parameter; obj = objectness exponent; mk = `min_keep`; rpn = RPN test top-n. Caps of 100 boxes per class and per image apply from v3 on.

Version	Thr	σ	obj	Stage box	Shift	mk	rpn	λ	Other
raw (model's own eval)	0.05			last	0		1000	0	
v1	1e-3	.75	.25	last	0		1000	1	per-class Gaussian soft-NMS
v2	1e-3	.75	.25	mean	-0.4		1000	1	
v3 (`pp_eval.FINAL`)	1e-5	.75	.25	mean	-0.4	30	1000	1	
v3′ (for `--flip-bias` runs)	1e-5	.75	.25	mean	0	30	1000	1	automatic in `pp_eval.py`
Submission, 10-27 to 10-31	1e-5	.75	.25	mean	-0.2	30	1000	1	F2, LR scale; BP scale has shift 0
e180 recipe, 11-01	1e-5	.5	0	mean	-0.2	30	2000	2	
56.49 recipe	1e-5	.5	0	mean	-0.2	30	2000	2	averaged weights; `--fusion agn --gamma 1 --topk 9`
FINAL	1e-5	.5	0	mean	-0.2	100	2000	2	56.49 recipe plus lattice snap

A.4. Run-name collisions

Names were reused during the project. These are the traps found in the sources.

Table 9. Run-name collisions.

Name	Meanings
`V2`	NWD on IN21k (parked, never run), and the 120-epoch 2-GPU run `V2_ddp`
`F1`, `F2`	Parked folds, and the full-data models. F1 also means macro-F1
`S5`	Mona on E09b (staging, never run), and the spatial prior screen `S5_sp`
`D3` = `W3_dino_spml`	The folder name of D3 starts with W3, which is unrelated to `W3`, the averaged local weights e70–120 of Part C
`E00`	Folder `ref_s0/vit_pretrained_lora`
`BP7`	`BP7_spatial64` (LoRA r32, head 1024, 36 epochs, 7,512 images, `$RUNS/BP7_spatial64`). From 10-29 on, "BP7" means LoRA r80, α160, head 1536, SpatialPrior64
45.83 (rounded)	Two different entries: SC2 e80 (45.8331) and BP7 local lr1x e130
`TWIN1`, `TWIN2`	TWIN1 is seed 0 of the final recipe. TWIN2 is the GIoU + IoU-branch variant, not a seed. `RPT_TWIN_cos200_s1` and `_s2` are seeds 1 and 2
`v2pp` in zip names	`V2_e100_val_v2pp.zip` and `V2_e120_val_v2pp.zip` use post-processing v3
`pp_eval.FINAL`	Is v3, not the FINAL of Table 8
A, B, C	Context trio submissions (§D.8); "candidate A" on 11-01 is the e100–180 window; A to D are also kNN fusion modes (§E.12)
L, R1, R2	Scale L versus runs L36, L1, L2. Screens R1, R2 versus R01, a reference seed that never ran

B. Timeline and who did what

Two lines ran in parallel from the TA's reference repository.

Table 10. The two lines of work.

Line	Scope
Rayner line	Branch `origin/rayner`, RTX 3080 on another machine, 10-12 to 10-23. Holds all work dated before 10-20 and the whole YOLO family (§C.1). Also tooling: submission preparation, early stopping, metrics and box visualisers. No run folders on this server, so all numbers come from `git show e9969d3:docs/dino_yolo.md`. An idea list in `docs/improvement.md` on this branch (10-13) names LoRA on MLPs, higher rank, EMA, threshold 0.001, 4–6 px anchors, soft-NMS, Cascade and retraining on all data. Several later won on the other line; the logs do not show whether that work read this file
Junkai line	Branch `junkai` (formerly `junkai-claude`) on the shared server with 2 RTX 6000 Ada. First `events.log` line 10-20 06:05. Everything from E00 through F2, SC1, SC2, BP7, cos200, all post-processing, lattice snapping, the twin protocol and the runs for this report. Code and execution were done with AI coding assistants directed by the team: Claude Code until 10-28, Codex from 10-28 to 11-01, Claude Code again from the evening of 11-01 to 11-02 and from 11-05. The claim in an older repository note that "experiments ran 10-20 to 10-23" is true of this line only

Table 11. Project timeline (SGT). Run details are in the sections below.

When	Event
10-12 to 10-13	Rayner line: CNN spatial prior configs, submission and analysis tooling
10-20 00:50	Rayner line: DINOv3 + YOLO family starts
10-20 06:05	Junkai line: first run event. E00 reference, then backbone and adaptation screens
10-20 07:07–07:50	Post-processing v1, without retraining
10-20 to 10-21	Cascade, NWD, longer schedules; V2 to 120 epochs; post-processing v2 and v3; V2 e100 scored 38.23 (10-20 22:23 by session timestamp); F1 scored 41.26 (10-21 04:54); F2 trained
10-22 00:57 to 10-23 15:42	Automatic fallback queue runs 36 seed repeats of C36. GPU work stopped on 10-23 at 15:54
10-27 16:11	F2 shift -0.2 recorded at 44.84 on the new leaderboard
10-27 17:32–20:11	SC1 trains; scored 28.90 at 20:53
10-28 02:07–08:28	SC2_full150 on 2 GPUs. Scored at 09:35: e130 51.86, e150 51.81
10-28 09:44 to 10-29 08:31	Bridge and playground screens BP1–BP5, BP7–BP9
10-29 14:49–20:57	BP7 full, 130-epoch cosine; paired previews scored. At 21:07 the team stops architecture work and turns to LR and epochs
10-29 22:04 to 10-30 21:30	lr1x, lr05x, hybrid130; lr125x stopped at e69
10-30 21:31 to 10-31 03:45	Local cos160, stopped at e154. CPU trials on cached outputs at about the same time
10-31 03:34–12:09	TIDE and data audit; classification-only context (trained 03:48–11:46, submitted 11:50–12:01); backbone audit
10-31 12:17–19:16	BP7 full, cos160, stop at e130: **52.78**
10-31 19:41	cos200 launched; three interruptions follow (§E.1). LingBot trains alongside and is paused at e64 on 11-01 at about 00:59
11-01 11:17–20:59	Thirteen official BP7 submissions. **56.49 posted 20:59:42**
11-01 21:09–21:48	AUG1 and DP1 screens: failed or killed, then stopped when the team changed the protocol
11-01 21:49	TWIN1 starts on the vallike protocol
11-01 about 23:00	Annotation lattice found; `snap_submission.py` written
11-01 23:59	Phase 1 closes. Final rank 3 of 9
11-02 00:01–00:05	Test set downloaded; candidates and the FINAL zip built automatically
11-02 00:11–00:27	Local kNN label-transfer experiment; team declines to use it
11-02 12:02–12:20	TWIN1 finishes at e190 and is evaluated. TWIN2 starts 12:04
11-02 21:28	Team stops all experiments. TWIN2 killed in e145; last checkpoint e140
11-05	Report work resumes. Runs for this report start at 11:49:56

C. Part A, 10-12 to 10-27: first work up to F2

C.1. Rayner's YOLO line

All numbers in this subsection are from repository documents only[‡], on scale Ry. Commit dates are from `git log origin/rayner`.

Table 12. Rayner line. Results are test Δm on scale Ry.

Date	Commit	Idea and why	Implementation	Result	Decision
10-12	`447367a`	CNN spatial prior for small objects	`sarbench/model.py`, `run_all_3080.sh`: 4 adapter configs with and without prior, batch 8 for a 10 GB card	vit + MoE-LoRA + prior, 24 ep: 92.8 / 28.6, +4.5. No with/without table in the documents; batch 8 differs from protocol	None recorded
10-13	`2ca51ab` `8f203ca` `27e22cd` `8a224cc`	Tooling	Submission zip preparation, early stopping (`--patience`), metrics and manual box visualisers	n/a	Kept as tools

(continued)

Date	Commit	Idea and why	Implementation	Result	Decision
by 10-20 00:50	runs `full_seed0`, `single_seed0*`	Longer training of TerraMind + MoE-LoRA	Epochs 24 / 50 / 100	+5.0 / +16.5 / +21.8	Early evidence that epochs dominate
10-20 00:50	`dea971d` `cbd32e5` `d84efcd`	Second family: fuse DINOv3 features into a YOLO	`sarbench/dino_yolo.py`, `train_yolo.py`; YOLO12-s + ViT-S/16, 31.2M all trainable, ViT at 0.1× LR, zero-init 1×1 fusion, mosaic and others	24 ep +7.4; 50 ep +15.0; 100 ep +19.0. `--cls-head` at 24 ep: F1 +3.0/+4.1, mAP −1.9/−1.4, test +6.9	Class head off by default
10-20 10:42–18:11	`fea565b` `4d91512` `a0d936d`	YOLO26 and a plain baseline	`--yolo yolo26s`, `--dino none`, `yolo26s-p2`	Plain YOLO26-s +1.4; DINO + YOLO26-s +9.4; P2 +7.6; DINO + YOLO26-s 100 ep +21.3 (3.9 h)	P2 not better (mAP −1.1/−1.2, 1.5 h vs 0.9 h)
10-21 07:57	`a1e920f`	90° rotations against small data	`--rot90 0.75`, `RandomRot90`	mAP −3.1/−2.7, Δm −6.4/−5.2 (YOLO12/26). Untested hypothesis in the document: radar shadow and layover direction	Off by default
10-21 21:56	`cda7acc`	Frozen ViT-B/16 + LoRA, so a bigger ViT fits	`--dino-adapt frozen|lora`	Frozen +2.1; LoRA r16 +8.7; LoRA on MLPs too +8.3	"LoRA on MLPs changes nothing"; contrast §C.3
10-22 06:55	`bfac2af`	LoRA rank sweep	r16 / 64 / 256, qkv + proj	+8.7 / +7.8 / +10.0	See seed note, §4.1
10-22 22:14	`e85cdfd`	Partial unfreezing and YOLO26-m	`--dino-unfreeze K`	Last 2 blocks +5.5; last 4 +7.0; LoRA + unfreeze 2 +9.2; YOLO26-m +9.2	Below r256
10-23 12:10	`e9969d3`	Longest ViT-B run	ViT-B/16 + LoRA r256 + YOLO26-s, 100 ep, 8.7 h	Test +21.2, val +22.5	"Level with ViT-S" at 3.5× total parameters

One further point exists: YOLO11-s "improved (48 ep)" 93.08 / 36.93, official +20.53. Its only source is a later team draft from 10-29; there is no code, run or date in this repository[‡]. The Rayner documents also give a seed note: "$\pm 0.9\ \Delta m$, trust only differences above about 1.5"[‡].

C.2. Baseline and infrastructure

Table 13. Baseline and infrastructure on the Junkai line, 10-20.

Item	What	Numbers
E00 `ref_s0/vit_pretrained_lora`	Reference recipe: ViT-B/16 IN21k + LoRA r16 attention only, 24 ep, flips, seed 0, local train 7,512. Purpose: a local Δm reference, since split and data amount differ from the official ones. Only one seed, unlike the official 3-seed reference; the planned R01 never ran	107,039,257 total, 20,997,913 trainable. Raw val 91.50 / 22.41 / AP50 48.63, test 91.61 / 23.31 / 52.77. v1 val 92.45 / 23.70, test 92.67 / 25.12, L +3.93. 1.02 h
Budget guard	`train.py` exits if parameters + float buffers exceed 130M or trainable exceed 40M; `--allow-over-budget` is only for reference runs	Lines 25, 372 of the F2 snapshot of `train.py`
Benches A00, A01	A00: reference 54.5 img/s, DINOv3 all-linear LoRA 44.4; first compile result unusable because compile time was counted. A01: 2-epoch steady state, compile +12 % with the same losses. `--compile` on for ViT runs	37.0 → 41.5; `events.log` shows 37.0 and 41.9
Two jobs on one GPU	Tested, then one job per GPU	23.4 + 23.5 vs about 52 img/s alone[†]
Queue	Per-GPU runners, `watchdog.sh` (backlog, else a new-seed replicate of `fallback_args.txt`), `events.log`, HOLD jobs for DDP	`$RUNS/queue/`
Code review	Automated adversarial review with 13 agents; 6 confirmed bugs fixed in `7afacd7`, `bba17ee`: `_orig_mod` keys broke loading compiled runs; `--train-norms` missed ConvNeXt LayerNorm2d (E03 crash); `predict.py` defaults were not the final post-processing; NWD via `cdist` lost precision under TF32; Cascade appended GT rows at each stage (1×/2×/3×); copy-paste carried neighbours' pixels	E00–E02 and E09b finished or started before the fixes

C.3. Backbone and adaptation

All runs: 24 epochs, local train, seed 0. L is the mean of local val and test.

Table 14. Backbone and adaptation screens (scale L). M = trainable/total parameters in millions.

Run	Hypothesis; flags vs E00	M	h	Raw val F1/mAP	Raw test F1/mAP	Raw L	v1 L	Decision and stated reason
E01 `E01_dino_ref`	Better pretraining. `--backbone dinov3`	21.0/ 106.25	1.06	93.42/ 25.49	93.07/ 26.80	8.10	10.88	Keep backbone. By size, DINOv3 is weak below 16 px and strong above†
E02 `E02_vit_all32`	Is adaptation capacity the limit? IN21k, LoRA r32 α64 on qkv, proj, fc1, fc2, `--train-norms`, `--backbone-lr 3e-4`	24.87/ 110.87	1.13	92.92/ 26.70	93.85/ 28.52	11.37	14.91	Stepping stone: more than the backbone swap alone
E03 `p10_E03_cnx_all32`	ConvNeXt-B LoRA for stride-4 features. `--backbone convnext_dinov3`, LoRA on `mlp.fc1,fc2`	n/a	sec	crash				Runtime error in `NativeLayerNorm` backward: the stem LayerNorm had only its bias trainable (bug above). Fixed job parked; **never run**
E09b `E09b_dino_wide64`	Literature "wide LoRA" recipe plus both gains. r64, α64, `--loraplus 2 --drop-path 0.1 --train-bias --train-ls --wd-hygiene`	29.66/ 114.80	1.04	95.23/ 24.36	94.74/ 25.30	6.18	9.07	Dropped: best F1, detection fell. Dependent jobs held. Cause never isolated (§K)
E07 `E07_cnxT_full`	ConvNeXt-T full fine-tune fits 40M; reference shows full FT $\gg$ LoRA. `--backbone convnext_t_dinov3 --adapt full --box-head small --drop-path 0.1 --backbone-lr 5e-5 --wd-hygiene`, no compile	35.48/ 35.48	0.89	94.66/ 24.98	93.82/ 25.59	6.78	9.09	Dropped: below E02, though most parameter-efficient. AP below 8 px 1.9, worse than reference 4.0†; weakness attributed to DINOv3 pretraining
E04r `E04r_dino_all32`	Diagnostic: DINOv3 + E02's plain recipe. E02 flags on DINOv3	24.87/ 110.08	1.00	94.77/ 29.65	94.85/ 30.98	18.08	21.40	**New base**; val and test equal at v1
D6 `D6_r64`	Rank 64. E04r, r64 α128	29.59/ 114.80	1.05	94.11/ 30.70	94.57/ 30.94	18.95	22.03	Dropped: gain too small for +4.7M. Test mAP v1 fell 32.52→32.03, val rose 31.10→32.17

Never run: E09c (E09b without drop path), E05, E05b, E06, E08, E10, E04 (cancelled), R01. All sit in `$RUNS/queue/parked` or `staging`, none in `done`. DINOv2 ViT-B/14 was rejected on paper because patch 14 misaligns the grid at 512†.

C.4. Post-processing v1

Done without retraining on 10-20 between 07:07 and 07:50.

Table 15. Post-processing v1 steps.

Step	Measurement	Decision
Threshold 0.05 → 0.001 on E00	Val 22.41 → 22.56, test 23.30 → 23.43 (`ref_s0/pp_study.log`)	Kept
Re-score boxes with image-class probability (`mult`, `agn`, `hard`)	Within ±0.05 mAP on E00, read as: errors are localisation, not class	Rejected at this point. Class-agnostic fusion returned on 11-01 (§E.6)
Detection-informed classification λ	E00 F1 val/test: λ0 91.50/91.61, λ0.5 92.15/92.33, λ1 92.45/92.67, λ2 91.98/93.21. On strong models ±0.2–0.6†; on 3 long models mean F1 at λ 0, 0.5, 1 = 95.02, 95.14, 95.22†	λ1 kept; integer to limit tuning
Soft-NMS σ 0.75 + objectness$^{0.25}$	E00 val 22.41 → 23.70, test 23.31 → 25.12, including λ and threshold. σ 0.5–1 and power 0.25–0.5 "about the same"†	Kept; first found by a research agent, reproduced by the main session

E00 under this post-processing is +0.84 before soft-NMS and +3.93 after it. These are two stages, not a conflict.

C.5. Detector for small objects

Base E04r, or an IN21k control. 24 epochs unless noted. A screening rule was set at this point: both holdouts improve and mean mAP gains at least 0.4.

Table 16. Detection head and small-object screens (scale L). M = trainable parameters in millions.

Run	Hypothesis, mechanism, flags	M	h	Raw val/test mAP	Raw L	v1 L	Decision
V1 `V1_cascade`	Cascade refines boxes; AP50 ≫ AP75. IN21k E02 + `--cascade --aligned-roi` (`sarbench/tiny.py`: 3 stages, class-agnostic light heads, weights 1 / 0.5 / 0.25)	23.80	1.30	28.34/ 30.18 (+1.64/ +1.66 vs E02)	15.59	17.97	Passes the rule; control
D1 `D1_cascade`	Same on DINOv3. E04r + `--cascade --aligned-roi`	23.80	1.28	31.04/ 32.21 (+1.39/ +1.23)	21.01	23.35	**Keep**
D2 `D2_nwd`	IoU is near 0 for an offset 8 px box; use Gaussian distance in RPN matching. `--rpn-nwd 8 --anchor-offset 0.5`, anchors `6,8,11;16,19,23;32,38,45;64,76,91;128,152,181`	24.87	1.12	30.53/ 31.55 (+0.88/ +0.57)	19.74	22.95	**Keep**
D3 `W3_dino_spml`	Spatial prior + multi-level neck to fix objects below 16 px. `--spatial-prior 64 --ml-levels 2,5,8,11`	25.08	1.19	24.36/ 24.67	5.14	9.04	Dropped. An older note blames the neck; no neck-only run exists (inference). Prior alone later: S5
D4 `D4_d4cp`	Rotations and copy-paste augment rare small objects. `--aug d4 --copy-paste 0.5`	24.87	1.05	24.90/ 25.80	7.13	9.83	Dropped. Copy-paste alone and D4 alone at 24–36 ep never run; D4 retested long in L1
D5 `p18_D5_mona`	Mona multi-scale adapters. `--mona 64 --compile`; 27.56M / 112.76M; exact no-op at init (2.68M, 24 adapters)	27.56	crash				cuDNN `BAD_PARAM` error in a depthwise conv of the compiled graph, first iteration. Not retried without `--compile`. Dropped; surer options available

Planned, not run: S1–S5 on E09b, V2–V5 on E02. GIoU loss (`--giou`, commit `938d493`) was implemented and appears only in A01; no scored run in this period has it. `--pretrained-prior` was never used.

C.6. Training length

Training length was the largest single gain of this period.

Table 17. Length screens (scale L, local train, seed 0).

Run	Change	h, peak GiB	Raw val/test mAP	Raw L	Post-processed L	Decision
L36 `L36_dino_36ep`	E04r, 24 → 36 ep	1.50, 42.6	34.91/36.03	29.35	v1 32.77; v3-equivalent 34.30	"Largest single gain"
C36 `C_casc_nwd_36ep`	E04r + Cascade + NWD, 36 ep	2.10, 42.6	35.34/37.25	31.28	v1 33.16; v3 35.46	Base for all later screens
C48 `C_casc_nwd_48ep`	Same, 48 ep	2.71, 42.6	39.33/39.85	38.62	v1 39.6†; v3 42.28	Longer is better on both holdouts

Aborted and not scored: `FINAL_v1` (36 ep full data, killed at e10) and `FINAL_v2` (48 ep, killed at e3), both on team request; `V2L_local` and `V2F_full` (two 1-GPU 120-epoch runs, killed after 2 minutes, because the team wanted a multi-GPU run).

V2 `v2_ddp`. Goal: run until growth stalls. C48 recipe, cosine 120 ep, train+val 8,452 images, local test 940 monitored every 10 epochs, 2-GPU DDP (`--train-split train+val --eval-splits test --eval-every 10 --save-every 10`, `NCCL_P2P_DISABLE=1`), 3.86 h, peak 37.95 GiB. The first DDP attempts hung in NCCL at DDP construction; PCIe P2P was the stated cause† and disabling it worked.

Table 18. V2 on local test. v2 and v3 columns recomputed from raw dumps. See §K on an older mislabelling of the v2 row as v3.

Epoch	10	20	30	40	50	60	70	80	90	100	110	120
Raw mAP	23.52	30.21	34.10	39.07	41.37	43.44	44.79	46.10	46.48	47.03	46.62	46.86
Raw F1	93.72	94.41	93.09	94.81	95.13	95.55	95.82	95.45	95.55	95.73	95.63	95.65
v2 mAP							46.92	47.84	48.31	48.91	48.61	48.54
v3 mAP							47.04	47.97	48.44	49.07	48.77	48.70

Decision: e100 chosen on the monitor set. Epochs 100–120 differ by about 0.4, within noise. Selection bias was noted at the time. The count of 100 epochs carried to F1 and F2 as a cosine ending at 100.

5-fold. `tools/make_folds.py` built folds (`86c1bf2`) and jobs `b0..b4_F*_c36` were queued. The team cancelled them in favour of new ideas, and they never ran.

C.7. Post-processing v2 and v3

Table 19. Post-processing v2 and v3 steps. Numbers are mAP.

Step	Why	Evidence	Decision
Shift −0.4 on 512 input	Research agent hint; tested independently on 3 models	C48 39.85→40.31, 40.28→40.95†. C36 re-score at shift 0 / −0.2 / −0.4 / −0.6: val 36.74 / 37.11 / 37.22 / 37.24, test 38.42 / 38.84 / 39.04 / 38.88	Kept (`6b9dab2`)
Separate corners, x/y, original-px shift	Is the offset uniform?	"Within ±0.02 of uniform −0.4" on C48†	No change
Stage-box mean (`stage_box='mean'`)	Cascade already averages class probabilities; same forward pass	C48 40.33→40.86, 40.97→41.51; C36 36.71→37.15, 38.51→38.98; V2 e70 46.58→46.92†. Stage weights differ by at most 0.06; small boxes on an earlier stage at most +0.04	Kept (`dd87834`)
Threshold 1e-5, at least 30 boxes per image	84 official val images had no box in the e100 zip; low-ranked boxes should add recall	V2 +0.16 (recomputed for e100), C48 +0.09, C36 +0.06†	Kept (`91f024d`)

C.8. 36-epoch screens against C36

All runs: 36 epochs, local train, seed 0, post-processing v3 or v3′. "vs mean" uses the 37-run C36 mean L 35.56 ± 0.705 (§4.1), with $z = \Delta/0.705$. "vs single" uses C36 seed 0, L 35.46, which is what was compared at decision time.

Table 20. Screens on C36. Every row is one seed.

Run	Hypothesis; flags	Val F1/mAP	Test F1/mAP	L	vs mean (z)	vs single	Decision and reason
S8 `S8_stride8`	Half of boxes are below 1 patch; overlapping patches double the token grid. `--token-stride 8`, stride-8 neck, DDP, 28 img/s	94.91/ 37.09	95.59/ 38.40	34.57	−0.99 (−1.4)	−0.89	**Dropped.** Early lead (e12 +1.29) faded to +0.44 at e24, +0.12 at e36 raw. 2.6× compute. Mean per-class vs C36: oil tank −6.8, harbor −4.6, bridge +0.7, playground +1.4, wind +1.6. Cause not investigated
E1 `E1_ema`	EMA usually helps long runs. `--ema 0.9998`	95.98/ 37.40	95.02/ 38.74	35.43	−0.13 (−0.2)	−0.03	**Dropped.** Last iterate 37.51/38.74 in the same run. e8–e24 lead of about +1 mAP vanished as LR→0
A4 `A4_sizeiou`	Small objects get no positives at stage IoU 0.6/0.7. `--cascade-size-iou`, table in `tiny.py:122`: below 8 px .40/.45/.50, 8–16 .45/.50/.55, 16–32 .50/.55/.65, else .5/.6/.7	95.27/ 38.15	95.27/ 39.78	37.25	+1.69 (+2.4)	+1.79	**Kept.** Trailed C36 to e16, led from e20
B1 `B1_iou`	Predicted IoU re-ranks boxes. `--iou-branch 0.5` (1,025 params), `quality_pow`	95.27/ 37.25	95.01/ 38.75	35.08	−0.48 (−0.7)	−0.38	**Dropped.** Rescoring power 0/1/2: val 37.25/36.88/36.40, test 38.75/38.47/38.09

(continued)

Run	Hypothesis; flags	Val F1/mAP	Test F1/mAP	L	vs mean (z)	vs single	Decision and reason
A2 `A2_flipfix`	Flips mirror annotation bias, so targets swing ± 0.3 px. `--flip-bias -0.3` (`data.py`: a flip moves boxes by $2b$, b in original px; `flip` aug only)	95.74/ 37.86	95.00/ 39.61	36.79	+1.23 (+1.75)	+1.33	**Kept.** Removes the need for -0.4. Shift sweep: 0: 37.86/39.61; -0.2: 38.04/39.80 (L 37.20, +1.65 vs mean); +0.2: 37.44/39.12
A4b `A4b_small16`	Protect 32–96 px by relaxing only below 16 px. `--size-iou-table small16`: below 8 .40/.45/.50, 8–16 .45/.50/.60	95.94/ 37.82	95.02/ 39.41	36.60	+1.04 (+1.5)	+1.14	**Dropped.** Below A4 by 0.33/0.37; 32–96 AP fell anyway
AA `AA_a4a2`	Do A4 and A2 stack? Local twin of F2. Both flags	95.05/ 38.43	95.55/ 40.01	37.82	+2.26 (+3.2)	+2.36	**Confirms the F2 combination after F2 ran** (AA started 14:09, F2 ended 12:02). Sub-additive. With -0.2 shift: 38.59/40.19, L 38.20, +2.64 (+3.75)
R1 `R1_nwd16`	Looser NWD for small boxes. `--rpn-nwd 16`	95.64/ 38.05	95.39/ 39.08	36.52	+0.96 (+1.4)	+1.06	Not adopted: test +0.04 vs single, within noise
R2 `R2_roi1024`	More RoI samples. `--roi-batch 1024`	95.91/ 35.38	95.62/ 37.08	31.54	-4.02 (-5.7)	-3.92	Not adopted; cause not investigated. Only clearly harmful screen; 2.50 h
S5 `S5_sp`	Spatial prior alone at longer schedule. `--spatial-prior 64`, random init	95.39/ 38.23	95.04/ 39.79	37.32	+1.76 (+2.5)	+1.86	Finished after F2; flagged as next candidate. Not in F2
S6 `S6_pe`	Adapt patch embedding to SAR. `--train-patch-embed --full-lr 5e-5`	94.98/ 37.62	95.23/ 39.39	36.16	+0.60 (+0.9)	+0.70	Not adopted, small
N1 `N1_top2full`	Spend budget on last 2 blocks. `--unfreeze-last 2 --full-lr 3e-5`, 37.19M / 108.23M	95.05/ 38.05	95.21/ 39.69	36.96	+1.40 (+2.0)	+1.50	Not in F2, finished after it; close to the 40M limit

Per-class means against C36. A4: wind turbine +6.5, oil tank +3.1, playground +1.2, ship +1.2, airport -1.6, aircraft -0.9, car -0.8, harbor -0.7. AA: wind turbine +10.6, oil tank +3.3, playground +1.2, aircraft -2.0, airport -1.6, harbor -1.0.

Table 21. AP by box side at 512 input, mean of local val and test, v3 (recomputed).

	all	<8	8–16	16–32	32–96	≥ 96
C36	38.1	8.2	17.0	28.4	52.4	47.6
A4	39.0	12.9	20.7	28.2	50.4	47.6
A4b	38.6	12.5	21.3	29.6	49.8	47.8

The explanation given for the 32–96 drop is that extra low-IoU positives of small objects affect the shared classifier. It is inference and was not tested. A4 and A2 are single-seed; the seed-0 C36 sits below the mean on val (37.22 vs 37.46), so gains stated at decision time differ from gains against the mean. Val mAP deltas against the mean: A4 +0.69 (was +0.93), A2 +0.40 (was +0.64). Test: A4 +0.87 (was +0.74), A2 +0.70 (was +0.57).

C.9. Long augmentation screens

Cosine 100 epochs, train+val, local test, 1 GPU. Added because augmentation might need long training to pay.

Table 22. Long augmentation screens. Baseline: V2 e100 v3 49.07, e120 v3 48.70.

Run	Flags	h	Raw mAP by epoch	Final v3 F1/mAP	LT	Decision
L1 `L1_d4long`	`--aug d4`	6.15	e10 20.43 … e90 43.60, final 43.70	95.89 / 45.43	49.80	Dropped; behind V2 (raw) at every epoch
L2 `L2_zoomlong`	`--zoom-p 0.3` (max 1.5)	6.30	e10 22.06 … e90 45.51, final 45.67	95.68 / 47.68	54.51	Dropped

Sessions compared these with 48.91 and about 48.5, which are v2 numbers. The conclusion stands and the gap is larger on v3.

C.10. Submissions up to F2

Table 23. Submissions up to F2 (files in `submissions/`).

Zip	Model, post-processing	Boxes	Local	Official
`interim_E02_val.zip`, `interim_E04r_val.zip`	Safety backups	101,757 / n/a		Not scored
`V2_e70_val.zip`	V2 e70, v2. GPU paused 33 s	54,311	LT 52.9	No score recorded
`V2_e100_val.zip`	V2 e100, **v2** (thr 1e-3)	39,485	v2 48.91 / F1 95.59, LT 57.1	**38.2349**, F1 95.0593, mAP 45.6234, AP50 70.3755, acc 95.3717. Codabench. 84 images with no box
`V2_e100_val_v2pp.zip`, `V2_e120_val_v2pp.zip`	v3	73,244 / 70,509		Not scored; fallback only
`F1_full_val.zip`	V2 recipe, cosine 100 ep, all 9,392, v3. 3.54 h	76,758	None	**41.2598**, F1 95.9015, mAP 46.9684, AP50 71.5669, acc 96.2132. Codabench
`F2_full_val.zip`	F1 + A4 + A2, v3′ shift 0	79,992	None	Not scored, so the -0.2 effect on the official set is unmeasured
`F2_full_val_shift-0.2.zip`	`predict.py --from-raw --shift -0.2`, 12,363,762 bytes	79,969	None; AA twin with -0.2: 38.59/40.19	**44.84**, F1 95.46, mAP 48.97, AP50 71.53, acc 95.79. New leaderboard

F2 command: `$BK44/run/a3a1_F2_full_ddp.sh`, with `F2_extra_args.txt = --flip-bias -0.3`. F2 has 109,014,752 total and 23,805,152 trainable parameters, trained in 3.59 h with peak 37.95 GiB.

Gains are confounded on the official scale. V2 e100 $\rightarrow$ F1 (+3.03) changes three things: 11 % more data, a cosine ending at the used epoch, and post-processing v2 $\rightarrow$ v3. F1 $\rightarrow$ F2 (+3.58: mAP +2.00, F1 -0.44, AP50 about the same) changes A4, A2, shift $-0.4 \rightarrow -0.2$ and retraining noise. No single step was scored alone. The prediction before scoring was +2 to +3[†].

D. Part B, 10-27 to 10-31: F2 44.84 to BP7 52.78

Every run in this part is seed 0. "Post-processing" below means soft-NMS σ 0.75, objectness$^{0.25}$, three-stage box mean, score threshold 1e-5, min_keep 30, per-class 100, total 100, λ 1, with the shift stated. The later 11-01 settings are in Part C.

D.1. Scaling to budget: SC1 and SC2

Hypothesis. F2 used 23.8M of 40M trainable parameters, so widening existing modules could buy score. The team decided to keep the architecture, fill the budget, and test on the 80:20 split first, then on all data.

Implementation. `--lora-rank 80 --lora-alpha 160 --cascade-rep 1536`, plus `--grad-ckpt` only to fit shared GPUs. A candidate table was computed on CPU (`$BK28/REPORT.md` §1). `--unfreeze-last` was rejected because it adds no parameters. Parameters: 124,866,272 total, 39,656,672 trainable.

Table 24. SC1 against AA (36 ep, 7,512 images, seed 0, post-processing, shift 0, scale BP).

	Val F1 / mAP	Test F1 / mAP	Local Δm
AA	95.05 / 38.43	95.55 / 40.01	37.83
SC1	95.65 / 40.96	95.05 / 42.87	43.75

SC1 minus AA is +5.92 (+5.97 val, +5.87 test). By size, mean of holdouts: below 8 px +5.6, 8–16 +6.3, 16–32 +3.4, 32–96 +3.2, at least 96 0.0. Shift -0.2 on SC1 gives val 41.21, test 43.17, i.e. +0.25 / +0.30 mAP. Official: 28.90, F1 94.69, mAP 40.85, AP50 67.64; not posted. An AA leaderboard score of "about 24" is an explicit estimate, since AA was never submitted.

SC2. All 9,392 images, 150-epoch cosine, 2-GPU DDP, global batch 16, seed 0; exact command in `$RUNS/SC2_full150/run.sh`. The team asked for about 150 epochs so that the best checkpoint could be picked; a cosine depends on total epochs and could not be extended mid-run.

Table 25. SC2 against F2 (official val).

	Δm	F1	mAP	AP50	Acc
F2 (100 ep, 23.8M)	44.84	95.46	48.97	71.53	95.79
SC2 e130	**51.86**	95.35	52.66	73.18	95.58
SC2 e150	51.81	95.44	52.61	73.24	95.72

e130 was picked by official score; the 0.05 gap is called noise. The +7.02 over F2 mixes scaling and 100 $\rightarrow$ 150 epochs and cannot be split. Head-only and LoRA-only ablations of the scaling were proposed and never run. SC2 stayed the best entry until 10-31. Aborted attempts: `SC2_full_aborted_e8` (100 ep, restarted to 150 at team request), a 150-epoch attempt that died at e1 with its parent session, `SC2_full150_1gpu_aborted_e22` (stopped for the 2-GPU run), and `SC1s1_aborted` (seed-1 repeat, stopped at e1 by the team to spend the GPUs on full data).

D.2. Bridge and playground screens

Why. On AA these classes sat at AP 1.7 and 3.7, and raising both was estimated as a large gain. The team asked for a literature search before training, on one GPU. Common config: LoRA r32, α64, head 1024, 36 ep, 7,512 images, seed 0, batch 16, A4 + flip-bias -0.3, NWD 8, `--grad-ckpt`. Commands: `$RUNS/BP_queue/run_all.sh`, `run_queue2.sh`, `followup_20261028/queue.log`.

Table 26. Small-config screens (scale BP; class AP is the mean of val and test, difference to AA in brackets).

Run	Flags	Local Δm	Val/test mAP	Bridge	Playgr.	Other class moves vs AA	Decision
AA_a4a2	Baseline	37.83	38.43/ 40.01	1.70	3.72		Baseline
BP1_ctx3	`--roi-context 3` (shared)	40.08	39.45/ 40.84	2.54 (+0.84)	4.49 (+0.77)	Oil +3.14, airport +2.71, wind -1.07	Not carried as is; see BP8
BP2_clsw3	`--cls-weights 3:3,7:3`	38.29	38.19/ 40.42	2.60 (+0.90)	6.49 (+2.77)	Wind -2.09, harbor -1.52, oil +1.22	Dropped
top-n 2000	`predict.py --rpn-topn 2000` on AA	n/r	38.44/ 39.98	1.69	3.73		Dropped: no change
top-n 3000	`--rpn-topn 3000` on AA	n/r	38.44/ 39.97	1.67	3.70		Dropped
BP4_partial	`--partial-neg`	37.94	38.52/ 40.12	2.27 (+0.57)	4.78 (+1.06)	Test F1 -0.85	Dropped
BP4 + fusion	Image-class probability $\times$ score	39.25	39.19/ 40.65	2.10	4.90		Dropped
BP5_partial_ctx3	`--partial-neg --roi-context 3`	39.14	38.91/ 40.76	2.31 (+0.61)	4.31 (+0.58)	Airport +3.34, harbor +2.26, ship -1.26	Dropped
BP5 + fusion		40.63	39.71/ 41.33	2.32	4.38		Dropped
BP3_os2	`--oversample 2:2,6:2`	33.68	36.72/ 37.73	2.33 (+0.63)	7.06 (+3.33)	Wind -9.97, harbor -4.70, airport -4.67, oil -3.49	Dropped
BP7_spatial64	`--spatial-prior 64`	**41.06**	40.02/ 41.37	2.32 (+0.62)	5.25 (+1.53)	Airport +3.60, oil +3.11, harbor +2.28, car +1.20	**Moved to full scale**
BP8_ctx3_cls_only	`--roi-context 3 --roi-context-mode classification`	40.86	39.82/ 41.33	2.29 (+0.59)	5.13 (+1.41)	Oil +5.35, airport +1.97, aircraft +1.57	Retested big (§D.8)
BP9_ctx3_clsw3	`--roi-context 3 --cls-weights 3:3,7:3`	39.19	39.34/ 40.31	2.60 (+0.91)	6.31 (+2.59)	Wind -2.10	Dropped

Mechanisms (`sarbench/tiny.py`, `train.py`).

- *RoI context:* `context_boxes()` enlarges each RoI, each side = $\max(3 \times \text{side}, 32\,\text{px})$. `LightBoxHead` adds a second 1×1 `ctx_reduce` conv: 49,344 parameters over 3 stages.
- *Classification mode:* the same head runs twice. `feat = head(local, ctx)` feeds classification and `box_feat = head(local)` feeds regression.
- `partial_negative_ce`: in a class-c image an unmatched RoI learns only $-\log(1-p_c)$ rather than "background".
- `--cls-weights`: class weights in the RoI cross-entropy. `--oversample`: `WeightedRandomSampler`, same epoch length, single GPU only.

- `--spatial-prior 64`: `SpatialPrior` CNN on raw pixels at strides 4 and 8, added through zero-initialised 1×1 convs into `SimpleFeaturePyramid` (`sarbench/model.py`). 210,688 parameters (derived: 24,015,840 − 23,805,152). Random initialisation; `--pretrained-prior` not used.

Fusion check on AA†. Multiplying by image-class probability gave mAP val/test 38.49–38.51 / 40.04–40.08, with bridge and playground unchanged. Dropped at that point. Class-agnostic fusion returned on 11-01.

Table 27. Error analysis on AA. Recorded only in session notes†; no artifact found.

	AP at IoU 0.1	RPN recall (IoU $\geq$ 0.5)	Median detector class probability on matched boxes
Bridge	17	53 %	0.08
Playground	18	67 %	0.06
Wind turbine	97	96 %	0.97

Table 28. Positive-sample diagnostic (AA model, 16 fixed train images per class, 144 in total; a sampling diagnostic, not an AP). Tool `tools/diagnose_roi_samples.py`, output `roi_diagnostic.json`.

Class	Stage 1/2/3 positives	GT covered excl. appended GT, stage 1/2/3	CE neg/pos ratio
Bridge	296 / 323 / 283	0.76 / 0.72 / 0.60	0.31 / 0.36 / 0.39
Playground	274 / 338 / 328	0.96 / 0.96 / 0.81	0.41 / 0.42 / 0.38
Wind turbine	1373 / 1681 / 1764	1.00 / 1.00 / 1.00	0.98 / 0.89 / 0.86

The diagnostic shows fewer positives for the two classes, and that negative loss does *not* dominate. This contradicts the earlier framing that background swamps them.

Proposed, never run: BP6 (down-weight high-confidence negatives), self-training pseudo-labels, EQL v2, Seesaw, dynamic or separately normalised positive/negative loss. None of these screens was submitted to the board in small config. Bridge and playground moved by at most +0.91 and +3.33 AP, and the largest playground gain cost about 4 local Δm (BP3).

D.3. Spatial prior on the big model

Hypothesis. S5 was cheap and fits the 0.34M of remaining budget. Config (`$EXP/bp7_full130/README.md`): SC2 + `--spatial-prior 64`, 9,392 images, 130-epoch cosine, DDP, per-rank workers 2 (SC2 used 8). 125,076,960 total, 39,867,360 trainable. 6.13 h; rank-0 peak 11.77 GiB.

Table 29. Paired stage previews on official val (Δm / mAP / F1).

Epoch	BP7 cos130	SC2 cos150	BP7−SC2 Δm
e40	32.89 / 42.68 / 95.61	30.70 / 41.45 / 95.88	+2.19
e80	48.40 / 50.75 / 95.71	45.83 / 49.37 / 95.86	+2.57
e110	51.60 / 52.45 / 95.65	not scored	
e130	51.76 / 52.53 / 95.65 (AP50 73.32)	51.86 / 52.66 / 95.35	**−0.10**

Confounds stated in the run's README: 130 vs 150 cosine, workers 2 vs 8, one seed, same trajectory across epochs. At e80 the LR was 3.27e-5 for BP7 vs 4.53e-5 for SC2†. At e130 of a 150-epoch plan the LR factor is about 4.4 % (derived), while BP7 cos130 at e130 is at 0, so the −0.10 does not isolate the prior. SC2 stayed best, but the BP7 architecture was kept as the base for schedule work because the team chose to stop changing architecture.

D.4. Learning-rate search

Why. LR had never been searched, and the team chose to stop architecture changes and optimise LR and epochs. Fixed BP7, local 7,512 images, 130 epochs.

Added support. `--schedule-epochs`, `--resume`, `--stop-after-epoch`, `--reproducible-loader`, `--save-training-state`; `sarbench/training_state.py` (full state with AdamW, scheduler, per-rank RNG, compatibility signature); `tools/evaluate_lr_checkpoint.py` (local eval with submission post-processing).

Pre-fixed rule. Metric = local val Δm at e130, submission post-processing. Within 1.0, prefer lr1x, then hybrid, then lr05x. The 1.0 is operational, not statistical. A planned local-test confirmation never ran because the queue was superseded.

Table 30. LR search (scale LR: local val, shift −0.2, seed 0).

Run	Head / backbone LR	Δm	F1	mAP	AP50	Bridge	Playgr.	Train h	Decision
lr1x	1e-4 / 3e-4	**68.70**	95.32	**52.27**	73.12	4.78	22.94	5.79	Kept
lr05x	5e-5 / 1.5e-4	61.49 (−7.21)	95.54	48.99	72.45	3.52	17.23	6.03	Dropped
hybrid130	1e-4 / 2e-4	65.92 (−2.77)	95.23	51.05	72.18	4.52	20.37	9.31 (shared)	Dropped
lr125x	1.25e-4 / 3.75e-4	no final eval						stop e69	Dropped by team

Table 31. Raw curves on local val: mAP (F1).

Epoch	lr1x	lr05x	hybrid	lr125x	cos160
10	22.24 (93.51)	21.14 (92.11)	23.78 (93.18)	21.94 (92.93)	22.91 (92.83)
30	32.35 (93.35)	35.97 (95.34)	37.51 (95.56)	32.05 (93.99)	32.26 (92.97)
60	45.61 (96.10)	44.35 (95.21)	45.41 (95.96)	43.88 (95.41)	43.83 (95.26)
100	51.05 (95.33)	47.74 (95.48)	50.02 (95.47)		49.74 (95.14)
120	51.49 (95.33)	47.90 (95.52)	50.09 (95.39)		51.51 (95.01)
final / e130	51.43 (95.24)	47.89 (95.43)	50.19 (95.39)		51.99 (94.92)

Early leaders at e30 (lr05x +3.62, hybrid +5.16 mAP over lr1x) finish behind. The assistant's prior had favoured hybrid; that prior lost. lr1x official: 45.83, F1 94.80, mAP 49.66, AP50 72.07, acc 95.09. lr125x was stopped at e69 with mAP 1.73 behind lr1x at e60, when the team chose to leave LR unchanged and try a longer schedule. The verdict on +25 % therefore rests on a partial curve.

D.5. Slower cosine

Hypothesis. LR reaches 0 too quickly, so tail gains are limited. Under warm-up then cosine, changing the horizon changes the LR at every epoch (`lr_factor` in `train.py`). Comparing 130- and 160-epoch plans is therefore not "30 extra epochs".

Local cos160 (`$EXP/bp7_fixed160/README.md`). Only `--epochs` and `--schedule-epochs` differ from lr1x. Stopped at e154 by the team; no e160 and no post-processed e130 local file exist.

Table 32. Pair at e130, 7,512 images.

	Official	Raw local val
cos130 (lr1x)	Δm 45.83, F1 94.80, mAP 49.66, AP50 72.07	mAP 51.43, F1 95.24, AP50 73.25
cos160	Δm 47.12, F1 94.58, mAP 50.41, AP50 71.62	mAP 51.99, F1 94.92, AP50 73.14
Change	+1.29, F1 −0.22, mAP +0.75, AP50 −0.45	mAP +0.56, F1 −0.32, AP50 −0.11

Later cos160 checkpoints were not better on the board: e130 47.12, e150 46.92. Post-processed cos160 e150: local val Δm 69.60, mAP 52.63, F1 95.51; local test Δm 65.80, mAP 53.23, F1 94.57. Update counts: local130 = 61,100; local160 = 75,200; full130 = 76,310; full160 = 93,920. Local 160 and full 130 differ by about 1.5 %. An auto-queue (`tools/run_bp7_full_selected.py`) would have selected 160 only if more than 1.0 better locally and then trained on all data; it was cancelled before it ran.

Full cos160, stop at e130 (`$EXP/bp7_full_cos160_stop130/README.md`, `tools/run_bp7_full_stop130.py`). `--epochs 160 --schedule-epochs 160 --stop-after-epoch 130 --reproducible-loader --save-training-state`, BP7 architecture, seed 0, 9,392 images, 76,310 updates. LR at e130: 8.53 % of peak. e130 was pre-specified as the sole candidate.

Table 33. Full-data models at e130 (official val).

Model	Δm	F1	mAP	Acc	AP50
BP7 cos130 e130	51.76	95.65	52.53	95.93	73.32
SC2 cos150 e130	51.86	95.35	52.66	95.58	73.18
BP7 cos160 e130	**52.78**	95.35	**53.15**	95.58	**73.76**

Differences: +1.02 vs BP7 cos130, +0.92 vs SC2. The +1.02 is not a pure schedule effect: BP7 cos130 lacked `--reproducible-loader` and cos160 had it. An interim preview (BP7 cos160 e50, `aaa.zip`, 121,485 boxes) has no score recorded anywhere.

Next schedule choices. Whether no LR decay would be better was raised, and constant LR to e130 versus a WSD tail from a shared e110 was proposed; the literature was mixed. With one day left the team chose the likely best option without control groups: cos200 with stops at e130 and e160, sharing GPUs with LingBot (`tools/run_bp7_cos200_final.py`, `tools/train_gpu_limited.py`, 20 % per-process allocator cap). Constant LR and WSD were never run. Details of cos200 are in §E.1.

D.6. CPU post-processing trials

All on `$RUNS/SC1_r80h1536/raw_val.pt` (940 images, CPU), shift −0.2. Baseline val mAP 41.2098, AP50 68.0708, AP75 43.0989, F1 95.6468, 78,365 boxes.

Table 34. CPU trials on cached SC1 outputs.

Trial	Hypothesis	Mechanism	Result (val mAP)	Decision
Box voting `tools/evaluate_box_voting.py`	Pre-NMS same-class candidates can refine coordinates	IoU $\{0.8, 0.9\} \times \alpha$ $\{0.10, 0.25\}$, weighted mean of same-forward candidates, scores unchanged	41.2040 / 41.1810 / 41.1993 / 41.2027, all below baseline (−0.0058 to −0.0288)	Dropped; test not opened
Small-box voting `tools/evaluate_size_gated_voting.py`	Only small boxes gain	$\sqrt{\text{area}} < 16$ or < 32	41.2036 / 41.2015	Dropped; exploratory
Remove total-100 cap `tools/evaluate_detection_cap.py`	603 of 940 images hit the cap, 345 multi-class; wrong classes may crowd out right ones	`total=None`, per-class 100 kept	41.2073 (−0.0025), boxes 78,365 → 129,283	Dropped

Official check of voting: `$SUB/SC2_e130_val_shift-0.2_boxvote_iou0.8_alpha0.10.zip`. The team noted almost no difference; no number was recorded.

Table 35. Proposal coverage on SC1 local val: GT with any RPN proposal at IoU ≥ 0.5 (not recall or AP). `proposal_coverage.json`.

Group	GT	Top 500	Top 1000	Top 1000 at IoU ≥ 0.75
All	3,323	88.44 %	89.77 %	60.58 %
Below 16 px	1,698	80.74 %	82.51 %	47.82 %
Bridge	357	45.94 %	48.74 %	10.64 %
Playground	276	56.88 %	66.30 %	n/r

SC2 RPN 2000 (`tools/make_rpn2000_submission.py`): RPN test pre- and post-NMS 1000 → 2000 on SC2 e130. 61,896 boxes vs 61,683; 0 classification labels changed; 53.2 ms/image on a shared GPU; 0.96 GiB peak. No official score recorded for this file.

D.7. TIDE error decomposition

`$EXP/autoresearch_20261031/`, tool `tools/research_bp7_errors.py`, CPU only. TIDE 1.0.1 on `lr1x/predictions_val.json`, native post-processing. AP matches COCO exactly.

Table 36. TIDE on lr1x. The oracle dAP values are headroom; they are neither additive nor achievable gains.

IoU	AP	Miss dAP	Loc dAP	Bkg dAP
0.50	73.246	14.736	3.461	1.341
0.75	56.319	9.323	20.437	0.741

Candidates ranked by that note: (A) classification-only context on BP7, *run*, §D.8; (B) NWD ranking assignment, not run; (C) Cascade GIoU, not run in this period (`--giou` was then unsupported in Cascade mode); (D) step-aligned schedules (partly done as cos200), `--wd-hygiene` (not run), small post-processing calibration (done on 11-01). The old EMA 0.9998 same-run comparison was −0.221 val and −0.024 test mAP against the last iterate.

D.8. Classification-only context and its failure to transfer

Hypothesis. BP8 had positive small-config evidence, and bridge–playground confusion is half of image errors: 24 of 48 in lr1x. The team stopped local cos160 at e154 for this and cancelled the full queue.

Config. `$EXP/bp7_context160/README.md`: BP7 + `--roi-context 3 --roi-context-mode classification`, cos160, 7,512 images, seed 0. 125,126,304 total, 39,916,704 trainable (83,296 left). 7.93 h. Initialisation RNG consumption differs from baseline. Pairing used e150, since the baseline stopped at e154. After the positive val result the team asked for a second-holdout check and for three board submissions.

Table 37. Context, paired at e150. Local Δm on scale LR (val) and the same form on local test.

Holdout or board	Metric	Base cos160 e150	Context e150	Difference
Local val	Δm	69.60	71.86	**+2.26**
	mAP	52.63	53.60	+0.97
	F1	95.51	95.69	+0.19
Local test	Δm	65.80	65.69	**−0.12**
	mAP	53.23	53.05	−0.18
	F1	94.57	95.06	+0.49
Official	Δm	46.92 (A)	46.89 (B)	**−0.03**
	mAP	50.32	50.19	−0.13
	F1	94.48	94.89	+0.41
	AP50	71.65	72.07	+0.42

Other official context points: e130 (C) 46.93, F1 94.78, mAP 50.24, AP50 72.33, i.e. −0.19 against the no-context e130 47.12; e160 46.80, F1 94.89, mAP 50.15, AP50 72.02 (local val 72.00, mAP 53.69, F1 95.60).

Table 38. Per-class AP, context minus base, e150.

Split	Aircraft	Airport	Bridge	Car	Harbor	Oil	Playgr.	Ship	Wind
Val	+0.34	+0.22	**−0.38**	+1.46	**+3.66**	+0.79	+1.40	+1.64	−0.45
Test	+1.91	**−4.02**	**−0.24**	−1.16	**−1.33**	+1.92	+0.31	+1.15	−0.14

The local gain was mainly harbor, ship and car, about 78 % of the net val gain†. Bridge did not improve on either holdout, so the original aim was not met. Image classification went from 892 to 895 of 940 correct†; `accuracy` 94.89 → 95.21. **Decision:** context dropped and the cheaper plain BP7 kept. The sources say this is not proof that context is harmful. Raw mAP at e130 also favoured context on val (52.95 vs 51.99), and the official pair at e130 was −0.17 mAP.

D.9. Backbone audit

`$EXP/backbone_audit_20261031/`. Public-source and meta-device parameter counts only; no training.

Table 39. Backbone audit. * = backbone plus adapter plus same-size SC2 heads, not integrated end to end.

Candidate	Total with heads	Trainable	Status
SC2 DINOv3-B gray, r80	124,866,272	39,656,672	Reconstruction matches `metrics.json`
LingBot-B gray, r80	124,894,688	39,656,672	Trialled, next subsection
TIPSv2-B/14 RGB, r80	125,934,560*	39,656,672*	Not run
RADIOv3-B timm-512 RGB, r80	126,074,336*	39,655,136*	Not run; NVIDIA original 137.87M exceeds budget
DINOv3 ConvNeXt-B, MLP r80 / r64	127,116,192 / 124,228,512	39,594,016 / 36,706,336	Not run
ConvNeXtV2-B	127,242,528	39,594,016	Not run

Screened out with reasons in the README: PE-Spatial, EVA-02, SAM2.1 Hiera-B+, SwinV2, VMamba, MambaVision, InternImage, Franca-B, TerraMind/DOFA. DINOSAR was excluded because UniSAR-7M includes SAR detection and classification datasets. DINOv3-L/SAT and larger are over budget. No official "DINOv4" release was found.

D.10. LingBot

Config (`$EXP/lingbot_full_cos160_stop130/README.md`, `sarbench/lingbot.py`, `sarbench/vendor/lingbot_vision`, `tools/run_lingbot_full_stop130.py`). Only the backbone changes from BP7 (`--backbone lingbot_b`): LoRA80, head 1536, SP64, 9,392 images, cos160, planned stop e130. 125,105,376 total, 39,867,360 trainable, 27,664 floating buffers. Gray patch kernel = sum of RGB kernel; RoPE random coordinate scaling off.

Zero-training splice, `bbb.zip`. At the team's request, 88 head tensors (28,032,480 parameters) were copied from BP7 cos160 e50, with 0 training steps. Classification agreement with `aaa.zip` was 26.37 %, and predicted class counts shifted heavily. No official score is recorded for `bbb.zip` or `aaa.zip`.

Trained run. The first queue failed at first backward in the smoke test: per-block `torch.compile` with non-reentrant gradient checkpointing. `--compile` was removed and the run restarted. e60 official: **39.23**, F1 94.02, mAP 46.44, AP50 72.63, acc 94.60.

Decision. The team paused it to leave one BP7 job running: stopped at e64, step 37,568, resumable. No same-epoch, same-schedule comparison exists. The nearest references are BP7 cos130 e40 32.89 and e80 48.40, on a different schedule.

D.11. Official scores recorded in this period

Table 40. Official scores 10-27 to 11-01 00:53, in order. Images = training images. The file `official_scores_phase1.json` omits the 10-29 previews, so it is not a complete list.

SGT	Submission	Images	Δm	F1	mAP	AP50	Acc
10-27	F2 shift −0.2	9,392	44.84	95.46	48.97	71.53	95.79
10-27	SC1	7,512	28.90	94.69	40.85	67.64	
10-28	SC2 e130	9,392	51.86	95.35	52.66	73.18	95.58
10-28	SC2 e150	9,392	51.81	95.44	52.61	73.24	95.72
10-29	BP7 cos130 e40	9,392	32.89	95.61	42.68	70.72	95.93
10-29	SC2 e40	9,392	30.70	95.88	41.45	70.00	96.14
10-29	BP7 cos130 e80	9,392	48.40	95.71	50.75	74.00	96.00
10-29	SC2 e80	9,392	45.83	95.86	49.37	72.78	96.07
10-29	BP7 cos130 e110	9,392	51.60	95.65	52.45	73.48	95.93
10-29	BP7 cos130 e130	9,392	51.76	95.65	52.53	73.32	95.93
10-30	BP7 lr1x cos130 e130	7,512	45.83	94.80	49.66	72.07	95.09
10-30	SC2 e130 box voting	9,392	almost no difference; no number recorded				
10-31	BP7 cos160 e130	7,512	47.12	94.58	50.41	71.62	94.95
10-31	Context cos160 e160	7,512	46.80	94.89	50.15	72.02	95.16
10-31	A: BP7 cos160 e150	7,512	46.92	94.48	50.32	71.65	94.88
10-31	B: context e150	7,512	46.89	94.89	50.19	72.07	95.16
10-31	C: context e130	7,512	46.93	94.78	50.24	72.33	95.09
10-31	**BP7 full cos160 e130**	9,392	**52.78**	95.35	53.15	73.76	95.58
11-01 00:53	LingBot e60	9,392	39.23	94.02	46.44	72.63	94.60

Generated with no recorded score: `aaa.zip`, `bbb.zip`, `SC2_e130_val_shift-0.2_rpn2000.zip`, SC2 e120 and e140.

E. Part C, 11-01 to 11-05: cos200, post-processing, lattice, Phase 2

Model recipe common to all BP7 cos200 work (`$BK56/run/args.json`, `$PH2/launch_twin.sh`). Backbone: `--backbone dinov3 --init pretrained --adapt lora --lora-targets attn.qkv,attn.proj,mlp.fc1, mlp.fc2 --lora-rank 80 --lora-alpha 160 --train-norms`. Heads: `--cascade --cascade-rep 1536 --aligned-roi --cascade-size-iou --spatial-prior 64`. Anchors and matching: `--rpn-nwd 8 --anchor-offset 0.5 --anchor-sizes=6,8,11;16,19,23;32,38,45;64,76,91;128,152,181`. Training: `--flip-bias -0.3 --grad-ckpt --backbone-lr 0.0003 --lr 0.0001 --weight-decay 0.05 --warmup-epochs 1`, batch 16 (8 per GPU), 2-GPU DDP, seed 0. Parameters: 125,076,960 total, 39,867,360 trainable, 16 float buffers.

E.1. cos200, its interruptions and fixes

Hypothesis. The slower cosine had already won once on full data (51.76 → 52.78, cos130 → cos160), so a still slower cosine should win again with no architecture change. With one day left the team chose to maximise the likely score without control

groups. **Config:** `--epochs 200 --schedule-epochs 200`, 9,392 images, 587 steps per epoch, checkpoints every 10, `--save-training-state`. Stops were planned at e130 and e160, then extended to e200.

Table 41. LR factor $0.5(1+\cos(\pi(e-1)/199))$ (derived; matches recorded e130 0.2755, e160 0.0964, e180 0.0247).

e100	e110	e120	e130	e140	e150	e160	e170	e180	e190	e200
0.504	0.425	0.349	0.276	0.208	0.148	0.096	0.055	0.025	0.006	0

Cumulative training hours: e130 9.90 h, e160 11.75 h, e180 12.97 h, e200 14.14 h.

Table 42. cos200 interruptions: three during training, plus one failure before training. Calendar span 10-31 19:41 to 11-01 17:12, about 21.5 h.

#	When	Symptom	Cause	Fix	Lost
1	10-31 19:51, e2 iter about 100	OOM: tried to allocate 2.25 GiB, GPU 1 had 3.92 GiB free	Memory fraction 0.2 cap: 7.53 GiB used of 8.88 GiB allowed, 2.34 GiB reserved but unallocated (fragmentation). The GPU also held 8.57 GiB (LingBot, according to session notes) and another process with 24.34 GiB	`PYTORCH_ALLOC_CONF=expandable_segments:True`; resumed from the e1 checkpoint	Under 1 epoch; about 1.75 h wall
2	11-01 07:29, right after e130	OOM: tried to allocate 8.38 GiB, about 15 GiB physically free, 8.87 GiB allowed	Per-process cap (fraction 0.2) too low for a resumed RPN allocation; physical memory was sufficient†	Fraction 0.35 (about 15.5 GiB per GPU); e131 validated first. Resumed 11:11	About 3.7 h idle; no training lost
3	11-01 12:15, end of e147	Log frozen while the controller reported "running"	DataLoader `resource_sharer` could not create `/tmp/pymp-*` (ENOSPC); ranks waited forever. Detected 13:07–13:12 (`stall_diagnosis.json`)	Private HDD `TMPDIR`; 300 s step-progress watchdog; 17 CPU tests. Resumed 13:19; e160 done 14:09	About 1.1 h idle
(4)	11-01 14:42	`ValueError: Expected exactly one --save`	A driver check required an explicit flag	Parse with `train.parse_args()`; 38 tests; `_v2` suite started 14:47; e200 done 17:11	No GPU time

Trajectory integrity. Strict resume kept model, AdamW, scheduler and two-rank RNG states (validation JSONs: `optimizer_and_rank_rng_saved`, `trained_tensors_match`). The nine source checkpoints come from four directories; paths and SHAs are in `$BK56/run/averaging_manifest.json`. The first 29 epochs shared GPUs with LingBot at 11 to 12 img/s per GPU, against 25 to 26 after LingBot paused†.

E.2. Every official submission of 11-01

All BP7 rows use `--shift -0.2` and stage box mean unless stated. rpn = `--rpn-topn`; obj = `--obj-pow`; σ = `--soft-sigma`; λ = `--cls-lambda`. Sources: `$BK56/run/official_scores_phase1.json`, exact values from `official_result.json` files where present, otherwise pasted by the team.

Table 43. All official submissions on 11-01 SGT.

#	SGT	Weights	Post-processing	Δm	F1	mAP	AP50	Acc	Boxes	Decision
0	00:53	LingBot-Vision-B e60 (BP7 heads, cos160, full)	default	39.23	94.02	46.44	72.63	94.60	n/c	Paused at e64
1	11:17	e130	rpn1000 obj.25 σ.75 λ1	52.76	95.04	53.22	73.65	95.37	n/c	Below prior best 52.78; continue
2	14:27	e160	same	53.41	95.02	53.56	73.41	95.37	60,355	New best
3	14:53	e160	+ λ2	53.4467	95.0964	53.5632	73.4094	95.4418	60,355	Kept. 4 labels change, detection bytes identical
4	16:19	e180	λ2	53.4703	95.0918	53.5768	73.3117	95.4418	58,599	Plateau noted
5	16:41	e180	`--stage-box last`	52.7957	95.0918	53.2242	73.0881	95.4418	52,958	**Rejected** (-0.6746)
6	17:04	e180	σ.5	53.7639	95.0918	53.7302	73.7589	95.4418	50,722	Kept (+0.2936)
7	17:19	e180	σ.5 obj0	54.8147	95.0918	54.2794	74.1149	95.4418	60,846	Kept (+1.0508)

(continued)

#	SGT	Weights	Post-processing	Δm	F1	mAP	AP50	Acc	Boxes	Decision
8	17:36	e180	+ rpn2000	54.83	95.09	54.29	74.09	95.44	62,865	Kept (neutral, +0.02)
9	17:40	e200	same	54.71	95.09	54.23	73.90	95.44	62,532	**Rejected** (-0.12 vs e180)
10	20:04	avg e160, 170, 180	same	55.26	95.01	54.53	74.21	95.37	63,756	Kept (+0.43 vs #8); 2 labels changed†
11	20:14	avg e100, 120, 140, 160	same	55.41	95.20	54.56	74.91	95.51	75,848	Kept (+0.15 vs #10; only +0.03 mAP)
12	20:40	avg e100..e180 (9)	+ `--fusion hard`	56.41	95.27	55.06	74.79	95.58	59,134	Kept (+1.00 vs #11; two changes at once)
13	20:59	same	`--fusion agn --gamma 1 --topk 9`	**56.4868**	95.2730	55.1025	74.8487	95.5820	59,711	**Posted** 20:59:42

Table 44. Zip files of Table 43, in `submissions/`. n/c above = box count not checked.

#	Zip
0	`lingb_e60.zip`
1, 2	`BP7_full_cos200_e130.zip`, `BP7_full_cos200_e160.zip`
3, 4	`BP7_e160_cls2.zip`, `BP7_e180_cls2.zip`
5, 6	`BP7_e180_last_cls2.zip`, `BP7_e180_soft05_cls2.zip`
7, 8	`BP7_e180_obj0_soft05_cls2.zip`, `BP7_e180_rpn2000_obj0_soft05_cls2.zip`
9	`BP7_e200_rpn2000_obj0_soft05_cls2.zip`
10	`BP7_avg160170180_rpn2000_obj0_soft05_cls2.zip`
11	`BP7_avg100120140160_rpn2000_obj0_soft05_cls2.zip`
12	`BP7_avg100-180x9_rpn2000_obj0_soft05_cls2_hardfuse.zip`
13	`BP7_avg100-180x9_rpn2000_obj0_soft05_cls2_agn_g1_k9.zip`. Classification CSV hash `51fbb4bce1`, identical to #12

Generated, never scored officially: `BP7_avg100-180x9_rpn2000_obj0_soft05_cls2.zip` (window without fusion, 71,802 boxes), `..._soft04_cls2.zip` (67,040), `BP7_avg100-200x11_rpn2000_obj0_soft05_cls2.zip` (69,741), the `avg100-200x11` and `avg100120140160` hard-fusion variants, `BP7_avg80-160x9_...`, `BP7_avg80-180x11_...`, `..._agn_g1_k2.zip`, `BP7_full_cos200_e180.zip`, `BP7_full_cos200_e200.zip` (58,390), `BP7_avg100-180x9_agn_SNAP_lattice.zip` (59,711), `BP7_avg100-180x9_agn_mk100_SNAP.zip` (103,701).

E.3. Shared post-processing pipeline

From `sarbench/postprocess.py::detections_from_raw`, `sarbench/tiny.py`, `predict.py`, `pp_eval.py`. Let p be the mean over the three stages of softmax outputs and p_{img} the image classifier. With `--stage-box mean`, $b = (b_{s1} + b_{s2} + b_{s3})/3$, then $b \leftarrow b +$ shift (-0.2 px on the 512 input), clipped to the image. Box scores by `--fusion`:

$$\text{none:}\quad s_c = p_c \qquad \text{mult:}\quad s_c = p_c\, p_{\text{img}}(c)^{\gamma}$$

$$\text{hard:}\quad s_c = 1 - p_{\text{bg}} \ \text{ for } c = \arg\max p_{\text{img}},\ \text{else } 0 \qquad \text{agn:}\quad s_c = (1 - p_{\text{bg}})\, p_{\text{img}}(c)^{\gamma} \quad (\gamma{=}1,\ \text{topk}{=}9)$$

Then $s_c \leftarrow s_c \cdot \text{obj}^{\text{obj_pow}}$ if obj_pow > 0; threshold $s > 10^{-5}$; the min_keep rule (§E.9). Gaussian soft-NMS per class picks the argmax repeatedly with $s_j \leftarrow s_j \exp(-\text{IoU}^2/\sigma)$. Caps: 100 per class, 100 per image. Classification: $\arg\max_c [\log p_{\text{img}}(c) + \lambda \log q(c)]$, with $q(c) = \max_{\text{proposals}} p_{\text{det}}(c) + 10^{-6}$ (`predict.py::image_class`).

E.4. Each post-processing idea

Classification fusion λ. A low-cost F1 gain. A local prior on two old models and two holdouts found $\lambda 0.5$ worst in all four, $\lambda 1$ best on BP7, $\lambda 2$ best on SC1, hence "keep $\lambda 1$, try $\lambda 2$ cheaply" (`$EXP/bp7_full_cos200_final/classification_lambda_prior.json`). Official: 53.41 $\rightarrow$ 53.4467 (F1 about +0.08, 4 labels changed, net about one more correct image). Kept. On W3 averaged weights $\lambda 2$ was not better locally (val $\lambda 1$ 95.148 vs $\lambda 2$ 94.956; test 95.105 vs 95.096), so its adoption rests on the official score alone.

More epochs. e160 → e180 with the same post-processing: Δm +0.0237, mAP +0.0136. e180 → e200 with best post-processing: −0.12. Lengthening stopped. It was noted that a plateau under fixed post-processing does not bound the model.

Last-stage boxes (rejected). The last stage is the most strictly trained. Replayed from the cached raw output; a CPU replay gave small ranking differences, so a GPU replay was used. Official: −0.6746 (mAP −0.3526, AP50 −0.2236). Local later, `agn_last` vs `agn_k9`: −0.306 [−0.477, −0.138] val, −0.418 [−0.605, −0.251] test.

Soft-NMS σ. Official σ 0.75 → 0.5: +0.2936 (mAP +0.1534, AP50 +0.4472). σ 0.35 was recommended twice and never scored officially, because the averaging request superseded it.

Table 45. σ variants on W3 under agn, mAP vs `agn_k9` (`boot2`, `boot5` logs).

Variant	Val	Test
σ0.4	+0.008 [−0.033, +0.049]	+0.034 [−0.013, +0.086], P .913
σ0.6	−0.052 [−0.099, −0.003]	−0.021 [−0.063, +0.020]
Per-class σ (dense 0.4 / sparse 0.4 / dense 0.35)	−0.033 / −0.011 / −0.012	−0.006 / +0.020 / −0.001

After snapping, σ0.4 vs the snapped base gave −0.004 (P .19) val, +0.017 (P .73) test, −0.016 (P .38) vallike (session tool output). σ stayed at 0.5.

Objectness exponent. The original score was $p_{\text{det}} \cdot \text{obj}^{0.25}$. Dropping it gave official +1.0508 (mAP +0.5492, AP50 +0.3561); local +0.36 val, +0.23 test. Mechanism evidence, unlabelled and cached†: candidates above 1e-5 went 199,557 → 297,238; min_keep fallback images 268 → 186; final boxes 50,722 → 60,846; images at the 100-box cap 158 → 244; only 26,130 boxes identical; median objectness of proposals with $p_{\text{det}} \geq 0.1$ was 3.83e-5 (fourth root about 0.079). The effect is not only re-ranking and cannot be decomposed without labels.

RPN 1000 → 2000. Pre- and post-NMS test limits both change (`--rpn-topn`); all 1,426 images have 2,000 proposals; 0 classification changes. Official +0.02, local −0.024 / +0.011. Kept as neutral, at twice the RoI compute.

E.5. Checkpoint parameter averaging and the window sweep

Hypothesis. Weights wander in a flat basin near the end of training; averaging the trajectory removes noise. The team asked to try combining checkpoints at 17:42, specifying that parameters be averaged, not predictions.

Implementation. $\bar{\theta} = \sum_e w_e \theta_e$ with equal weights, on the 234 saved trained tensors (39,867,360 values); accumulated in FP64, cast to FP32; loaded with `--weights`; one model, one forward pass (`tools/average_ckpts.py`, `$BK56/run/averaging_manifest.json`). Budget unchanged; frozen tensors unaffected. LoRA subtlety: $\overline{B}\,\overline{A} \neq \overline{BA}$. The measured relative Frobenius difference for e180/e200 was 0.00214 %, and a rank-160 exact form would give 136.9M / 51.7M, over budget†. Parameter distance: e180 → e200 0.619 %, e160 → e180 2.628 %†. Checks: 234 of 234 tensors exact in `average_validation.json`; backup re-averaging matched bitwise†.

What the official rows identify. #10 vs #8: +0.24 mAP, −0.08 F1. #11 vs #8: +0.27 mAP, +0.11 F1. #10 vs #11 differ by 0.03 mAP, and the 0.15 Δm gap comes from 0.19 F1 (2 to 3 images†). **The e100–180 window alone was never scored.** #12 changed window and fusion together. One estimate of the window effect, +0.27 mAP (sd 0.09), came from a normal posterior with priors N(0.10, 0.15) and N(0.12, 0.12)†. That is an inference, not a measurement.

Local window sweep. W-series on `$RUNS/BP7_LR130_local_s0_20261029_v2/lr1x` (7,512 images, cos130), rpn2000, obj0, σ.5, λ2, shift −0.2, no fusion. Full-run equivalents are matched by LR position (local e70 factor 0.445 vs full e100 0.504). Source: `$BK56/run/local_holdout_window_sweep.json`.

Table 46. Averaging window sweep on local weights.

Window (checkpoints)	Full equivalent	Val mAP	Test mAP	Val F1	Test F1
e130 single	e200	52.70	53.10	95.24	94.45
e100–130 (4)	e160–200	52.72	53.22	95.24	94.56
e70–100 (4)	e100–160	52.26	53.30	95.30	94.75
W3 e70–120 (6)	**e100–180**	52.62	53.57	95.20	94.66
e70–130 (7)	e100–200	52.71	53.54	95.20	94.66
e50–130 (9)	e80–200	52.13	53.31	95.48	94.87
e50–100 (6)	e80–160	51.53	52.97	95.48	94.96

The pattern reported at the time: the window must include the low-LR tail, and starting before about half the peak LR costs

strict-IoU mAP. Local val alone shows averaging not winning (W3 −0.08 vs single); test alone shows +0.47.

E.6. Hard fusion, then class-agnostic fusion, with bootstrap

Hypothesis. Each image has exactly one class. Boxes of another class in a correctly classified image are false positives for that class's PR curve, and they use up the 100-box slots. Evidence on candidate A: 1,206 images with other-class boxes, 15,677 such boxes, only 4 scoring above 0.3, 339 images at the cap†. Earlier negative signals: on an old BP7 model `mult` moved val mAP 40.017 → 40.005†, and hard filtering had been ranked low. The result reversed that view.

A proof sketch was recorded in session notes†: removing false positives from a class's list cannot lower that class's AP; agn is weakly better than hard on misclassified images; rescoring with $1 - p_{\text{bg}}$ has no monotonicity guarantee and is empirical. Any post-processing that only rescales one class's scores cannot change that class's AP, since AP_c depends only on within-class ordering†.

Official. Hard + window +1.00 (mAP +0.50, F1 +0.07). agn over hard +0.08 (mAP +0.04, F1 0, CSV identical). A prediction of +0.13 Δm for agn† scaled the local gain by second-class probability mass (0.54 % per image on official val vs 0.88 / 1.01 % local; total 7.75 across 1,426 images; 902 such boxes, 10 with score above 0.1).

Local bootstrap. W3, 500 image resamples, paired. Our COCO evaluator was validated against pycocotools to 4 decimals: 52.6223 / 52.8107 val, 53.5740 / 53.6188 test (`tools/boot_eval.py`, `$RPT/prior_evidence/boot/`, `$BK56/run/local_holdout_postprocess_bootstrap_*.json`).

Table 47. Fusion variants against hard fusion, mAP points on W3 (bootstrap mean [95 % CI]).

Variant vs hard	Val	P>0	Test	P>0	Full-set val / test
None	−0.174 [−0.373, +0.016]	.036	−0.035 [−0.274, +0.246]	.372	52.622 / 53.574
agn γ1 k9	**+0.091 [+0.019, +0.176]**	**.996**	**+0.151 [+0.015, +0.363]**	**.992**	**52.907 / 53.780**
agn k2	+0.092 [+0.020, +0.177]	.996	+0.131 [−0.005, +0.332]	.966	52.907 / 53.763
agn k3	+0.091 [+0.020, +0.177]	.996	+0.149 [+0.011, +0.361]	.992	52.907 / 53.777
agn γ.5 k2	+0.063 [−0.000, +0.134]	.972	+0.107 [−0.022, +0.323]	.904	52.877 / 53.734
agn k2 with fused probs	+0.007 [−0.033, +0.037]	.684	+0.164 [+0.014, +0.411]	.990	52.819 / 53.780
hard with λ2 fused label	−0.055 [−0.174, +0.008]	.124	+0.141 [−0.005, +0.393]	.962	52.750 / 53.756
mult γ1	−0.032 [−0.238, +0.172]	.366	−0.041 [−0.262, +0.206]	.376	52.762 / 53.570
hard σ.4	+0.001 [−0.042, +0.047]	.500	+0.029 [−0.022, +0.083]	.860	52.816 / 53.634
hard σ.6	−0.052 [−0.098, −0.003]	.018	−0.020 [−0.060, +0.021]	.176	52.756 / 53.593
hard shift −0.1	+0.006 [−0.122, +0.134]	.534	−0.095 [−0.230, +0.057]	.108	52.815 / 53.522
hard shift −0.3	−0.469 [−0.619, −0.330]	.000	−0.411 [−0.572, −0.242]	.000	52.335 / 53.197

agn vs none by bootstrap: +0.266 [+0.057, +0.479] val, +0.186 [−0.007, +0.366] test. By size (none / hard / agn; session tool output): val APs 46.06 / 46.30 / 46.35, APm 61.42 / 61.87 / 61.93, APl 64.24 / 59.89 / 61.55; test APs 45.20 / 45.36 / 45.39, APm 62.88 / 63.07 / 63.11, APl 60.99 / 55.30 / 55.92. So hard and agn lower large-object AP on both holdouts.

Decision on the last two submissions. At 20:45 an estimate favoured A + hard, with expected +0.36 and P(beat 55.41) about 75–80 %†. The actual change was +1.00. The planned second shot (55.41 weights + hard) was replaced by agn because its estimated chance of beating the new score fell to about 0.2 %†. The team submitted and posted agn at 20:59.

E.7. Rejected post-processing

W3 weights unless stated; values against `agn_k9`; 300 resamples. Source: `$RPT/prior_evidence/boot/boot{2,3,5}_*.log`; comparisons are session tool output. Absolute values are shown where only those were logged; base was 52.89 / 53.78.

Table 48. Post-processing rejected on local holdouts (mAP).

Idea	Flags, definition	Val	Test	Keep
γ 1.5, γ 2	`--gamma`	+0.005, +0.009	−0.037, −0.051	no
Score threshold 1e-3		−0.079 [−0.099, −0.057]	−0.097 [−0.124, −0.072]	no
Stage-box weights (.2,.3,.5)	`stage_weights` (opt-in, added this night)	+0.015	−0.005	no
(0,.5,.5)		−0.099	−0.149	no
(.25,.25,.5)		+0.005	+0.025	no
(.5,.3,.2)		−0.309	−0.357	no

(continued)

Idea	Flags, definition	Val	Test	Keep
(.25,.5,.25)		−0.055	−0.016	no
σ.4 + weights (.2,.3,.5)		+0.046 [−0.044, +0.148]	+0.042 [−0.076, +0.135]	no
Iterative refine 1×	`--extra-iters 1` (with snap, vs snapped base 53.111 / 54.142)	+0.037 [−0.063, +0.148]	−0.052 [−0.149, +0.026]	no
Iterative refine in prob avg	`--extra-iters 1 --extra-probs`	Full-set 52.68 (vs 52.89)	53.65 (vs 53.78)	no
Stage prob weights (.5,1,1.5)	`--prob-weights`	52.74	53.63	no
(0,1,1)		52.06	52.89	no
(0,0,1)		46.29	46.87	no
(1.5,1,.5), snapped		+0.061 [−0.019, +0.183], P .963	+0.041 [−0.058, +0.136], P .793	no
(1.25,1,.75), snapped		−0.004 [−0.049, +0.089]	+0.050 [−0.010, +0.117], P .950	no
Per-class corner-bias table	Fit on one holdout, apply to the other; `snap_eval2.py`	−0.162 [−0.345, +0.078]	+0.033 [−0.127, +0.183]	no; does not transfer
bias_int + snap vs snap		−0.045 [−0.198, +0.100]	−0.152 [−0.306, −0.003]	no
Per-class affine calibration	`box_bias.py`, cross-fit	not recorded	−0.006	no
Round w, h (wind, rest)	`wh_eval.py`, vs snap + wind	−0.096 [−0.145, −0.051]	−0.102 [−0.147, −0.059]	no
Round w, h (oil tank 512)		−0.083 [−0.128, −0.043]	−0.105 [−0.143, −0.072]	no
Oil tank lattice r_0 = .3 / .5	`lattice_eval.py`, vs snap	−0.023 / +0.004	about −0.008 / +0.001	no
Count/rank prior	$s' = s \cdot P_c(N \geq r)^\alpha$, α .25 / .5 / 1	−2.011 / −2.432 / −3.123	−1.840 / −2.256 / −3.118	no
Size-to-class prior	$\log p' = \log p + \beta[\log P(c\|\text{size}) - \log P(c)]$, β .5 / 1	F1 94.956 / 95.055 (one image); mAP −0.014 / −0.027	F1 unchanged; mAP +0.004 / −0.001	no
Weight "soup"	Average of base + AUG1 averaged weights	not run	not run	queued only

IoU profile (W3, agn; session tool output). Val at IoU .50 to .95 in steps of .05: 74.6, 73.7, 71.6, 68.8, 64.5, 58.2, 49.7, 37.4, 22.4, 8.1. Test: 75.0, 73.9, 72.3, 69.3, 64.9, 59.8, 50.9, 38.2, 23.3, 10.2. AP90 is 1.3 to 8.1 for aircraft, car and ship, and 30 to 58 for airport, harbor, oil tank and wind turbine.

Table 49. Per-class AP on W3, agn (val / test).

Aircraft	Airport	Bridge	Car	Harbor	Oil tank	Playground	Ship	Wind
57.1/57.4	75.6/81.3	4.2/6.2	63.8/58.9	66.8/69.2	72.6/71.2	22.3/18.9	43.3/43.2	70.5/77.6

Training-side options mentioned but not run on any full-scale schedule: SABL, DFL + high-resolution stride, DETR family, COCO-pretrained heads, a small IoU head on frozen BP7 (1,537 parameters). A survey concluded they were not worth switching to in the remaining 6 days; its citations were not independently verified[†].

E.8. Annotation-lattice snapping

Trigger. The team looked at other teams' public scores at 22:15. Team 6 had AP50 only +0.76 above ours but mAP +4.86, which reads as a strict-IoU gap. Checking training labels found integer coordinates. Table 50 is session tool output on all 32,533 training boxes.

Table 50. Lattice by (class, image px) cell. The last column is the share of images with at least 2 boxes that share one fractional offset.

Cell	n	`xywh` all integer	Images with shared fractional offset
Aircraft 224 / 256 / 512	312 / 72 / 2,148	100 %	100 % of 38 / 4 / 524
Car 512	4,089	100 %	100 % of 423
Bridge 256	336	100 %	100 % of 64
Harbor 256	781	100 %	100 % of 225
Oil tank 256	2,325	100 %	100 % of 475
Ship 256	1,484	100 %	100 % of 195

(continued)

Cell	n	`xywh` all integer	Images with shared fractional offset
Ship 336	5	100 %	100 % of 1
Ship 512	53	50.9 %	61.5 % of 13
Oil tank 512	3,255	11.2 %	29.4 % of 630
Wind turbine 512	8,625	w, h integer 51.1 %, x, y integer 19.7 %	**40.2 % of 819**
Bridge 512	3,648	0.4 %	0.5 % of 664
Harbor 512	1,869	0.0 %	2.3 % of 395
Airport 512	919	0.0 %	0.5 % of 209
Playground 512	2,610	0.0 % (w, h integer 1.6 %)	1.2 % of 516
Ship 273, 391	1, 1	0 %	n/a

Wind turbine GT concentration $R \geq 0.9$ holds in .46 / .47 of holdout images. Median corner errors of true positives (W3, val, original px): oil tank 512 .20, wind .23, oil tank 256 .33, harbor 512 .63, playground .63, aircraft 256 .86, bridge 512 1.16, harbor 256 1.14, car 1.42, airport 1.58, ship 256 1.54, bridge 256 1.65, aircraft 512 2.83.

Algorithm. `tools/snap_submission.py IN.zip images.json OUT.zip [--no-lattice] [--r0 0.5] [--smin 0.3] [--nmin 3]`. It edits only the written boxes in the zip. Scores, categories, order and CSV are untouched.

1. *Integer cells.* For category aircraft (1) or car (4) at any size, or bridge (3), harbor (5), oil tank (6), ship (8) when $\max(w_0, h_0) < 400$: $x_1, y_1, x_2, y_2 \leftarrow \text{round}(x), \text{round}(y), \text{round}(x + w), \text{round}(y + h)$, clipped so that $x_1 \leq w_0 - 1$, $x_2 \leq w_0$, $x_2 \geq x_1 + 1$, and likewise for y.
2. *Wind-turbine lattice (category 9), per image.* Take boxes with score ≥ 0.3; at least 3 are required. Compute the circular mean of fractional parts, $z = \text{mean}(\exp(2\pi i \,\text{frac}(v)))$, over $\{x_1, x_2\}$ and over $\{y_1, y_2\}$, which gives the offset $o = \arg z / 2\pi$ and concentration $R = |z|$.
3. Only if $\min(R_x, R_y) \geq 0.5$: $v \leftarrow \text{round}(v - o) + o$ for all wind boxes in the image, clipped likewise.

Fixes during development: the first version left 59 boxes outside the image, clipping left 32, and an edge fix got 0. Maximum corner move 0.998 px; mean move of changed boxes 0.487 px; 28,507 of 59,711 val boxes changed.

Table 51. Snapping results on W3 agn, local holdouts, mAP (session tool output).

Variant	Val	Test
Integer snap (round), full set	52.907 → 53.084 (+0.177)	53.780 → 54.089 (+0.309)
Bootstrap	+0.185 [+0.045, +0.337], P 1.000	+0.326 [+0.197, +0.459], P 1.000
Half-way ($\alpha = 0.5$)	+0.061	not printed
Wind R_0 = 0 / .3 / **.5** / .7, on top of snap	−0.266 / +0.046 / **+0.027** / +0.001	−0.192 / +0.034 / **+0.053** / +0.014
Wind R_0 = .5 bootstrap	+0.030 [+0.007, +0.060], P 1.000	+0.051 [+0.021, +0.084], P .995
Integer + wind, total	53.111	54.142
Per-class gains, integer snap	Aircraft .12, bridge .05, car .06, harbor .13, oil tank .89, ship .34	Aircraft .47, bridge .05, car .34, harbor .01, oil tank 1.02, ship .88

Merged estimates†: 992 vallike +0.31 [+0.17, +0.45], i.e. +0.59 Δm; 1,880 merged +0.28 [+0.18, +0.40], i.e. +0.54 Δm. On TWIN1 the same step gives +0.20 [+0.07, +0.36] on vallike (Table 65).

Status. The snapped val zip `BP7_avg100-180x9_agn_SNAP_lattice.zip` (sha `65dc8ae5...`) was offered at 23:00, one hour before Phase 1 closed. No score appears in any artifact or note. **Snap has no official validation of its own**: the Phase 2 score covers only the combined entry (§E.13). Predicted R on official val: 220 wind images with at least 3 confident boxes; R quantiles 10/25/50/75/90 = .07, .14, .32, .65, .81; share with $R \geq .5$ = 0.39. Local val predicted share .14, test .33; GT $R \geq .9$ share .46 / .47. The assumption that val and test share this lattice was never tested on official data.

E.9. min_keep

Code in `sarbench/postprocess.py`:

```
keep = torch.where(scores > score_thresh)[0]
if len(keep) < min_keep:  # every image has objects: keep its min_keep best boxes even when all score low
    keep = scores.topk(min(min_keep, len(scores))).indices
    score_thresh = 0.0  # (only this image) soft-NMS must not drop them again
```

`min_keep=30` was the default inside every earlier submission. Flag: `predict.py --min-keep 100`. Box counts: val 59,711 → 103,701, test 89,110 → 155,989.

Table 52. min_keep on W3, mAP vs `agn_k9`, unsnapped.

min_keep	Val	Test
0	−0.000	+0.000
60	+0.005 [+0.001, +0.014], P 1.0	+0.007 [+0.001, +0.023], P 1.0
100	**+0.011 [+0.006, +0.019], P 1.0**	**+0.015 [+0.007, +0.028], P 1.0**

After snapping: +0.013, +0.016, +0.016 on val, test and vallike, P 1.00 (session tool output). See §K on the claim that AP cannot drop.

E.10. Combined gain estimate

FINAL (min_keep 100 + integer + wind) against the plain 56.49 recipe on W3 (session tool output, `phase2_gain.py`). Bootstrap 95 % CI; P>0 = 1.000 everywhere; Δm uses the test slope.

Table 53. Combined gain of the two Phase 2 additions, mAP.

Set	Full-set Δ	Bootstrap mean	95 % CI	Δm (test slope)
Local val 940	+0.218	+0.232	[+0.092, +0.383]	+0.46 [+0.18, +0.75]
Local test 940	+0.378	+0.397	[+0.262, +0.532]	+0.78 [+0.51, +1.05]
Merged 1,880	+0.297	+0.307	[+0.208, +0.404]	+0.60 [+0.41, +0.79]
vallike 992	+0.325	+0.326	[+0.174, +0.475]	+0.64 [+0.34, +0.93]
Test-like reweighted	+0.297	+0.300	[+0.197, +0.405]	+0.59 [+0.39, +0.80]

This was widened at the time to +0.2 to +1.1 Δm using the 0.3 to 1.9 local-to-official ratio†.

E.11. Phase 2 protocol

Plan at 21:15 on 11-01 (`$PH2/PLAN.md`). AUG1: `--speckle 0.2 --gamma-jitter 0.29 --photo-p 0.5` on the exact local cos160 recipe (7,512 images, stop e150). Speckle is multiplicative Gamma(L, L) with sd $\sim U(0.02, 0.2)$, $L = 1/\text{sd}^2$; gamma is $I^{\exp(U(-.29,.29))}$; each with probability .5 (`sarbench/data.py`). Why: only flip augmentation so far and overfitting signals. Decision rule: at least +0.3 mAP on both holdouts with averaged e80..e150. It was never applied.

Table 54. AUG1 and DP1 attempts.

Attempt	Time	Result	Reason
1	21:09:23 to 21:20:36	rc=1, SIGABRT	NCCL collective timeout 600 s on ALLGATHER. The log filename blames compile
DP1	21:20:57 to about 21:22:55	Killed	The queue started it in error after attempt 1 failed
2	About 21:23 to 21:32	Killed	Eager mode still hung
3	21:38:07 to about 21:48	Stopped at e3	NCCL fix applied; the team changed the protocol

Root cause of the hang, from `nccl_test` logs: default rc=124, `NCCL_CUMEM_ENABLE=0` rc=124, `NCCL_P2P_DISABLE=1` rc=0. An IOU1 smoke test hit an OOM (tried 8.26 GiB, 10.63 GiB reported free) at 21:40. The DP1, IOU1 and SA1 queues gave no meaningful training.

Table 55. Local-to-official calibration of post-processing deltas, mAP (`prior_evidence/boot/proxy_*`).

Change	Official	Local val, full set	Local test, full set
σ .75 → .5	+0.153	+0.159	+0.157
obj .25 → 0	+0.549	+0.360	+0.226
rpn 1k → 2k	+0.010	−0.024	+0.011
Last vs mean	−0.353	−0.228	−0.484
agn vs hard	+0.040	+0.096	+0.162
r / MAE / sign agreement		.981 / .082 / .80	.853 / .116 / 1.00

Val-like reweighted: r .982 / MAE .086 on val, .834 / .122 on test, so reweighting did not improve agreement. Here $n = 5$ deltas, and the two largest (objectness, last stage) dominate the correlation.

The vallike split

The team asked for a held-out subset matching the official val mix, because local holdouts also go into final training. External SAR data was considered for this and rejected (§I). Built instead: `cs701_data/dataset/SARFact-Course-20K/vallike_split.json`, `classification/{vallike,trainvl}.csv`.

- **vallike** (992 images, 4,156 boxes): a subset of the old local val + test, with a per-cell target from official val's (predicted class, size) mix, seed 0. Class shares within 0.1 point of val. No official labels are used.
- **trainvl** (8,400 images, 28,377 boxes): the rest, 89.4 % of train.

Cell targets / available (session tool output): aircraft 256 4/11, 512 99/225; airport 130/130; bridge 256 10/45, 512 74/205; car 64/88; harbor 256 59/63, 512 24/209; oil tank 256 68/111, 512 38/135; playground 131/236; ship 256 106/230, 512 5/8; wind 180/184. vallike therefore has few 512 px harbors and many wind turbines and airports. The same W3 weights score about 2.4 mAP higher on it: snapped 55.909 (min_keep 30) vs merged uniform 53.461 (min_keep 100). The contributions of composition and near-duplicate share (.264 vs about .205 above .98) were not separated.

Consistency checks, including the weak six-model check

The check above (Table 55) is on post-processing deltas. A second check asked whether local measures rank *models* as the board does. Six 7,512-image models with official scores, post-processing of that time (`fusion=none`, σ.75, obj.25, rpn1000, min_keep 30). Source: `$RPT/prior_evidence/hist_eval.log`.

Table 56. Six-model historical check.

Model	Off. Δm	Off. mAP	Off. F1	Merged 1,880 mAP / F1	Merged val-like mAP	vallike 992 mAP / F1
cos130 e130	45.83	49.66	94.80	52.237 / 95.245	53.464	55.268 / 95.659
cos160 e130	47.12	50.41	94.58	52.788 / 94.980	53.894	55.610 / 96.406
cos160 e150	46.92	50.32	94.48	52.759 / 95.047	53.919	55.631 / 96.386
ctx e130	46.93	50.24	94.78	53.073 / 95.240	54.091	55.943 / 96.157
ctx e150	46.89	50.19	94.89	53.139 / 95.382	54.153	56.054 / 96.454
ctx e160	46.80	50.15	94.89	53.154 / 95.336	54.165	56.027 / 96.351

Pearson r with official, $n = 6$ (derived). mAP: merged uniform 0.651, merged val-like 0.691, vallike 0.517. F1: merged uniform 0.942, vallike -0.212. Proxy score vs Δm: merged 0.694, vallike 0.750. Local sets rank the context models above cos160, and the board does the opposite. Paired (session tool output): `ctx_e150` − `cos130_e130` on vallike is +0.792 mAP [+0.204, +1.394]. This check is weak: $n = 6$, and it shows vallike can mis-rank whole models. It was not cited when the protocol was chosen; the correlations of .98 and .85 quoted at the time belong to the post-processing deltas of Table 55.

TWIN1

Config (`$PH2/launch_twin.sh`): `--train-split trainvl --eval-splits vallike --epochs 200 --schedule-epochs 200 --stop-after-epoch 190 --save-every 5 --eval-every 10 --save-training-state --reproducible-loader --batch-size 16 --workers 2 --eval-batch-size 4`, `NCCL_P2P_DISABLE=1`, HDD `TMPDIR`, 2-GPU DDP, seed 0. 525 steps per epoch. Run `$RUNS/TWIN1_base_trainvl8400_cos200_s0_20261101`, 11-01 21:49:31 to 11-02 12:02:16, rc=0. With `--stop-after-epoch` training writes no `model.pt`, so e190 is scored from `ckpt_e190.pt`.

Table 57. TWIN1 with FINAL post-processing on vallike (`$PH2/eval_twin1/verdict.txt`, 400 resamples). Per-class order: aircraft, airport, bridge, car, harbor, oil tank, playground, ship, wind.

Weights	mAP	F1	Bootstrap vs e130 mAP	Per-class AP
e130	57.295	96.005	n/a	59.4, 79.9, 7.8, 62.2, 80.7, 77.6, 30.4, 43.0, 74.6
e160	58.273	96.343	+0.972 [+.585, +1.371]	59.7, 81.3, 8.4, 61.6, 82.2, 77.9, 32.5, 41.8, 78.9
e180	58.254	96.647	+0.983 [+.572, +1.380]	59.6, 81.0, 7.9, 61.3, 83.0, 77.5, 32.6, 42.3, 79.1
e190	58.131	96.647	+0.864 [+.464, +1.269]	59.3, 81.0, 8.0, 61.0, 82.9, 77.6, 32.5, 41.8, 79.0
avg140–190 s5	58.494	96.700	+1.221 [+.839, +1.615]	59.7, 81.2, 8.0, 61.7, 83.6, 78.1, 32.8, 42.3, 79.1
avg120–190 s5	58.654	96.573	+1.367 [+.995, +1.723]	60.1, 81.4, 8.2, 61.9, 83.4, 78.6, 33.0, 42.3, 79.0
avg100–180 s10 (the entry's window)	**58.812**	**96.528**	+1.534 [+1.153, +1.892]	60.7, 81.3, 8.1, 62.2, 83.8, 78.8, 33.0, 42.4, 79.2
avg100–180 s5	58.876	96.550	+1.607 [+1.227, +1.951]	60.8, 81.1, 8.2, 62.5, 83.3, 79.2, 33.0, 42.5, 79.2
avg100–190 s5	58.897	96.573	+1.614 [+1.258, +1.991]	60.8, 81.1, 8.2, 62.6, 83.6, 79.0, 33.0, 42.5, 79.3

(continued)

Weights	mAP	F1	Bootstrap vs e130 mAP	Per-class AP
avg80–180 s10	58.782	96.799	+1.527 [+1.175, +1.886]	61.2, 81.1, 7.4, 63.3, 83.1, 79.2, 32.5, 42.5, 78.6

Decision. The first shot needed no change; the window spread of about 0.1 was treated as noise†. `verdict.txt` compares each row only against e130. The window minus best single (e160) is +0.539 mAP; §F now gives its bootstrap interval.

Training-time vallike mAP (TWIN1, training defaults): e10 24.96, e20 31.65, e30 37.50, e40 41.07, e50 43.00, e60 44.63, e70 49.03, e80 50.44, e90 51.42, e100 53.01, e110 53.78, e120 54.54, e130 55.53, e140 55.75, e150 56.25, e160 56.13, e170 56.39, e180 56.28, e190 56.19.

TWIN2: stopped, nothing concluded

Why. Teams 6 and 9 gain at strict IoU and our AP90 is low. Run `$RUNS/TWIN2_giou3_iou1_trainvl8400_cos200_s0_20261102`, flags `--giou 3.0 --iou-branch 1.0`. GIoU was added for Cascade at 22:24 on 11-01 (previously asserted off): each stage has `loss_giou_s{t}` $= w_t \cdot 3\cdot$ GIoU loss on decoded positive boxes (`sarbench/tiny.py:270-280`). The IoU branch sits on the last stage: BCE to the IoU of the decoded box with GT, `loss_iou` weight 1 (`sarbench/tiny.py:281-291`). Parameters 125,078,497 / 39,868,897 (+1,537). Test-time `quality_pow` rescoring at 0.5, 1, 2 was to be evaluated and never ran; there is no `eval_twin2`.

Table 58. TWIN2 training-time vallike mAP (training-default post-processing: no snap, no agn, no averaging, no quality rescoring).

Epoch	10	20	30	40	50	60	70	80	90	100	110	120	130	140
TWIN2	26.01	33.82	38.27	39.07	42.28	47.26	50.15	51.04	52.04	53.12	54.13	55.01	55.68	55.79
−TWIN1	+1.05	+2.17	+0.77	−2.00	−0.72	+2.63	+1.12	+0.60	+0.62	+0.11	+0.35	+0.47	+0.15	+0.04
AP50 Δ	−0.01	+1.99	−0.50	−1.66	−0.96	+1.03	+0.26	+0.40	−0.35	+0.55	−0.19	−0.52	−0.36	−0.31

It was killed at e145 iter about 300 (11-02 21:28:46) when the team stopped all experiments; no reason is recorded. There is no verdict, and **no conclusion about GIoU + IoU branch is supported**. The e80 to e120 points were read at the time as a strict-IoU lead†; the later e130 and e140 points narrow it.

Chain never reached: `TWIN3_aug` (`--speckle .2 --gamma-jitter .29 --photo-p .5`), `TWIN4_smallanchors` (`--anchor-sizes=5,7,10;...`), `TWIN5_droppath01` (`--drop-path 0.1`). No full-data retrain followed.

E.12. kNN label transfer

Motivation. Near-duplicates strongly affect scores (§G.4). Raised as a rules grey zone at 23:00 on 11-01 and tried at the team's request on local data only, never in a submission. **Implementation** (session scratch scripts `knn_eval.py`, `knn_eval2.py`, `align.py`): database of 7,512 local train images, queries of 1,880 local holdouts; nearest neighbour by frozen DINOv3 ViT-B/16 embedding; best of 8 dihedral transforms plus integer translation by phase correlation; ρ = Pearson correlation on the overlap; transfer the GT boxes.

Table 59. Alignment quality (session tool output).

ρ	n	Share	Same class	Recall @.5 / .75 / .9	Precision @.5
$\geq .95$	588	31 %	1.00	.91 / .69 / .39	.95
.85–.95	85		1.00	.90 / .80 / .73	.96
.70–.85	70		1.00	.88 / .73 / .58	.88
.50–.70	122		.95	.67 / .45 / .24	.82
.30–.50	159		.87	.38 / .20 / .09	.50
$< .30$	856		.72	.04 / .01 / .00	.07

Table 60. Fusion results against FINAL (val 53.125, test 54.158), mAP.

Mode	Val	Test
A: top score, drop overlapping model boxes	−1.28 to −1.70	−3.25 to −3.91
B: replace coordinates	−1.98 to −2.87	−2.59 to −3.32
C: add only	−2.68 to −4.35	−4.88 to −6.11

(continued)

Mode	Val	Test
D: recall-only $\rho \geq .5$, s=.3 (add where no same-class model box has IoU $\geq .3$)	**+0.517 [+.230, +.899], P 1.00** (+67 boxes)	**+0.288 [+.139, +.517], P 1.00** (+33 boxes)
D, $\rho \geq .7$, s=.3	+0.162 [+.042, +.304] (+20)	+0.076 [+.019, +.167] (+11)
D, $\rho \geq .5$, s=.05	+0.196 [+.088, +.342]	+0.109 [+.050, +.185]
D, $\rho \geq .5$, s=1	-0.546 [-1.166, +.375]	-0.028 [$-.525$, +.632]

The losses in A, B and C come mostly from wind turbine (-16 to -27 AP), where the model's sub-pixel fit beats the duplicate annotation. Classification override fixes 1 to 2 and breaks 0 to 1 of 940. Gains under D: playground +2.6 / +1.5, oil tank +1.6 / +0.7, plus harbor and bridge. **Rule analysis**†: it needs training images plus features at test time, which count against "every weight it uses at test time" (4.9M headroom); a second frozen network conflicts with "one model"; it exploits train/test duplicates; and it would need TA approval. **Decision:** the team declined to ask the TA and dropped it.

E.13. Shot selection and what is known about Phase 2

Table 61. Automatic build on 11-02 (`$PH2/watch_test_release.sh`, `snap_after_watch.sh`, `final_after_watch.sh`).

Time	Output	Boxes	Size MB	SHA256
00:03	`PHASE2_shot1_avg100-180x9_rpn2000_obj0_soft05_cls2_agn_g1_k9.zip` (exact 56.49 recipe)	89,110	13.8	`799cfecb...`
00:03	`PHASE2_backup_..._hardfuse.zip`	88,203	13.7	`da97370c...`
00:04	`..._agn_g1_k9_SNAP.zip` (56.49 + snap)	89,110; 28,582 int, 11,813 wind	2.9	`e1700e69...`
00:04	`PHASE2_shot1_mk100.zip` (intermediate)	155,989	24.2	not noted
00:05	**`PHASE2_shot1_FINAL_avg100-180x9_agn_mk100_SNAP.zip`**	155,989; 62,708 int, 11,812 wind	4.8	`c90b0d5b...`

All pass `tools/check_submission.py`: 2,065 images, 0 outside. All three shot zips share classification CSV hash `27b69b2d18`. Exact commands are in §M.

Selection logic at the time. The 56.49 recipe, plus two additions tested only locally. Shot 1 was first planned to stay exact (`PLAN.md`, 21:15), then FINAL was preferred. TWIN1 gave no reason to change the window.

Table 62. What is recorded about the Phase 2 submission.

Question	Finding
Phase 2 score	Read from the public board on 11-05 18:59 SGT: Δm **61.25**, macro-F1 96.96, mAP 55.74, AP50 75.19, accuracy 97.09, posted 11-03 15:37 SGT, submission `s_lia6Xuj5jW`. Rank 3 of 8 at that time (Team 6 71.57, Team 9 71.40, Team 2 58.36); the board locks on 11-08. The saved board is `results/leaderboard_phase2.json`
Which file was uploaded?	**Not established from project records.** Upload was manual and no upload log exists. On 11-05 the team recalled the FINAL zip, as a recollection
Evidence pointing elsewhere	At 21:22–21:23 on 11-02 a screenshot showed a file picker on the plain-recipe zip (`PHASE2_shot1_avg100-180x9_..._k9.zip`), with a question whether that file had the 56.49 recipe. Five minutes later all experiments were stopped
One or two submissions used?	Unknown. The plain recipe could have been an earlier submission. No evidence either way
Was a second submission planned?	Planned, then not pursued in any recorded form. The plan was: winner of the twin chain → full-data retrain → shot 2 by 11-06. The chain and retrain never ran. No `PHASE2_*` zip was written after 11-02 00:05
Why was TWIN2 stopped?	Team instruction with no reason recorded
Grading weight of the phases	Not in the docs. What is posted at the deadline counts; if nothing is posted, the last successful submission counts

Other analyses of the same night. From the public leaderboard API (`GET /api/leaderboard?phase=N`): strict band $(10\,\text{mAP} - \text{AP50})/9$ was ours 52.9, Team 6 58.3, Team 9 58.8†; AP50 74.85 vs 75.61 and 78.99; F1 similar. Phase 2 board at 20:30 on 11-02: Team 6 71.57 (mAP 61.21, F1 96.11), Team 9 71.40 (61.08, 96.27). Our expected test Δm was calibrated at about 60 to 62 from this†. Arithmetic: the 56.49 recipe's F1 and mAP under the test reference give 59.10 Δm; FINAL was expected at about 59.7†.

F. Runs added for this report

`$RPT/queue.py` adds repeats of the final recipe and leave-one-out ablations, because the submitted model has one seed and most screens were single-seed against older bases. All runs use a frozen code copy in `$RPT/code`, the split `trainvl` (8,400) → `vallike` (992), and FINAL post-processing for scoring. Training was still in progress when this text was written, so only the design is described here; results appear in the two generated tables below. A smoke pass (`queue.py --smoke`: 64 images, 1 epoch) of all seed-0 variants, one twin and the reference finished before launch at 11-05 11:49:56.

F.1. Design

Twin seeds. Seeds 1 and 2 of the final recipe with `--epochs 200 --schedule-epochs 200 --stop-after-epoch 180 --save-every 10 --eval-every 20 --save-training-state`. Rationale: the final recipe at full length has a single seed (TWIN1), so the averaging and post-processing gains have no seed spread yet. After training, `queue.py` averages e100–180 with step 10 (`eval_tools/average_local.py`) and dumps raw vallike predictions for the average and for e130, e160, e180. `eval_tools/ablate_pp.py` then scores the cumulative ladder and the removals. These runs are not strict replicates of TWIN1:

Table 63. Seed 0 twin against seeds 1 and 2.

	TWIN1 (seed 0)	`RPT_TWIN_cos200_s{1,2}`
GPUs and batch	2-GPU DDP, 8 per GPU, global 16	One GPU, batch 16
Stop	e190 (averaging window still e100–180)	e180
Checkpoints	Every 5	Every 10
Eval during training	Every 10	Every 20
Workers	2	4

The spread between the three therefore mixes seed and parallelism. One GPU per twin leaves the second GPU free for the leave-one-out queue.

Leave-one-out. Each variant removes one part from the final recipe and trains a 36-epoch cosine (`--epochs 36 --schedule-epochs 36 --eval-every 36 --save-training-state`), scored on the last checkpoint without averaging. Seeds 0, 1, 2. 36 epochs is the length of the earlier C36 screens (SC1 at this length took 2.64 h on 7,512 images). Short runs rank parts rather than predict full-length gains, since §D.5 shows that some gains pay late.

Table 64. Leave-one-out variants in `queue.py`. Shift is the box shift used when scoring.

Variant	Change from the final recipe	Shift	Rationale
`base`	None	−0.2	Paired baseline for every other row
`noSP`	No `--spatial-prior`	−0.2	The prior's effect was never isolated on the board (§D.3)
`noSizeIoU`	No `--cascade-size-iou`	−0.2	A4 is single-seed on the small model
`noFlipBias`	No `--flip-bias`	−0.4	Without the fix the earlier best shift was −0.4 (§G.2). Scoring this variant at −0.2 would charge it for a mismatched shift. The row measures flip bias together with its shift
`noNWD`	No `--rpn-nwd`	−0.2	D2 predates every later change
`noCascade`	No `--cascade`, `--cascade-rep`, `--cascade-size-iou`; `--aligned-roi` stays	−0.2	Removes the largest head. Scored with the same FINAL settings
`noSmallAnchors`	No `--anchor-offset`, `--anchor-sizes` (reference anchors)	−0.2	D2 changed anchors and matching at once
`noTrainNorms`	No `--train-norms`	−0.2	Never ablated alone
`smallCapacity`	LoRA r32 α64, head 1024	−0.2	Scaling (SC1, SC2) was never split or repeated

Reference. `RPT_REF24_s0` trains the reference recipe (`--backbone vit --init pretrained --adapt lora`, 24 epochs, seed 0) on the same split, for a same-split reference point. `eval_tools/analyze_loo.py` loads its metrics but does not use them in Table 2, where Δm uses the official val references applied to vallike.

Queue. Two GPU lanes. GPU 0 runs twin seed 1 then seed 2, then shares the remaining list. GPU 1 takes the shared list in order: all seed-0 variants, the reference, all seed-1 variants, all seed-2 variants. A failed training resumes from `training_state.pt` with up to 3 retries, 5 minutes apart. A job named in `skip.txt` is skipped, so the n column of Table 2 can be below 3. Every run uses `--grad-ckpt`, `--reproducible-loader`, `--batch-size 16`, `--workers 4`, `--eval-batch-size 4`. The raw dump uses `--rpn-topn 2000 --obj-pow 0 --soft-sigma 0.5 --cls-lambda 2 --fusion`

`agn --gamma 1 --topk 9 --dump-raw`; min_keep 100 and snapping are applied afterwards in `ablate_pp.py`.

F.2. Results

The two tables below are generated by `report_final/gen_report.py` from `$RPT/results/`.

Results of the leave-one-out runs are in Table 2.

Results of the post-processing ablation are in Table 3.

F.3. Seed 0 detail of the post-processing ablation

`$RPT/results/pp_ablation_TWIN1_s0.json` is finished: TWIN1, on vallike, paired bootstrap with 200 image resamples. Δm uses val references, and its interval uses the slopes 1.9135 and 0.5358. Table 65 adds intervals and removals to the cumulative ladder, for this one seed only.

Table 65. TWIN1 (seed 0) on vallike. Upper block: cumulative ladder, each step against the row named in the first column. Lower block: FINAL with one thing changed, difference = variant minus FINAL. P is the bootstrap share of FINAL or the later step being better on mAP; it carries no meaning when the mAP difference is exactly 0 (λ rows).

Step	mAP	AP50	F1	Boxes	ΔmAP [95 %]	P	ΔΔm [95 %]
Cumulative ladder							
e180, σ.75, obj$^{\cdot 25}$, RPN 1k, λ1	57.00	76.89	96.46	43,035			
σ0.5 (vs previous)	57.15	77.24	96.46	36,841	+0.15 [+0.08, +0.19]	1.000	+0.29 [+0.15, +0.37]
Objectness power 0	57.56	77.33	96.46	45,851	+0.41 [+0.21, +0.70]	1.000	+0.79 [+0.40, +1.34]
RPN 2000	57.60	77.34	96.46	47,455	+0.04 [−0.01, +0.09]	.955	+0.07 [−0.01, +0.16]
λ2	57.60	77.34	96.65	47,455	+0.00 [+0.00, +0.00]	n/a	+0.10 [−0.14, +0.37]
Average e100–180	58.37	78.24	96.53	53,058	+0.78 [+0.50, +1.05]	1.000	+1.42 [+0.74, +2.11]
Hard fusion	58.53	78.06	96.53	42,579	+0.16 [−0.13, +0.35]	.855	+0.31 [−0.26, +0.68]
agn (vs hard)	58.59	78.14	96.53	43,034	+0.06 [+0.01, +0.11]	1.000	+0.11 [+0.03, +0.22]
min_keep 100 (vs agn)	58.61	78.15	96.53	66,912	+0.02 [+0.01, +0.02]	1.000	+0.03 [+0.02, +0.04]
Lattice snap = FINAL	**58.81**	78.19	96.53	66,912	+0.20 [+0.07, +0.36]	1.000	+0.39 [+0.13, +0.69]
FINAL with one change							
Without snap	58.61	78.15	96.53	66,912	−0.20 [−0.36, −0.07]	1.000	−0.39 [−0.69, −0.13]
min_keep 30	58.80	78.18	96.53	43,034	−0.02 [−0.02, −0.01]	1.000	−0.03 [−0.04, −0.02]
No fusion	58.76	78.30	96.53	65,254	−0.06 [−0.23, +0.19]	.680	−0.11 [−0.45, +0.37]
Hard fusion	58.74	78.11	96.53	64,777	−0.07 [−0.12, −0.02]	1.000	−0.13 [−0.23, −0.04]
σ0.75	58.69	77.76	96.53	73,505	−0.12 [−0.20, −0.03]	.995	−0.23 [−0.39, −0.07]
Objectness$^{\cdot 25}$	58.46	78.09	96.53	66,128	−0.36 [−0.56, −0.20]	1.000	−0.68 [−1.07, −0.38]
RPN 1000	58.82	78.21	96.54	72,657	+0.01 [−0.03, +0.04]	.430	+0.02 [−0.19, +0.19]
Last-stage box	58.45	78.36	96.53	58,188	−0.36 [−0.53, −0.21]	1.000	−0.69 [−1.02, −0.40]
Shift 0	58.14	78.14	96.53	66,921	−0.67 [−0.93, −0.41]	1.000	−1.29 [−1.78, −0.79]
Shift −0.4	57.69	78.28	96.53	66,907	−1.12 [−1.38, −0.81]	1.000	−2.14 [−2.65, −1.55]
λ1	58.81	78.19	96.53	66,912	+0.00 [+0.00, +0.00]	n/a	+0.00 [−0.37, +0.29]
Single e180	58.25	77.37	96.65	69,942	−0.56 [−0.77, −0.32]	1.000	−1.00 [−1.56, −0.47]
Single e160	58.27	77.52	96.34	69,579	−0.54 [−0.82, −0.29]	1.000	−1.13 [−1.88, −0.57]

Four readings of this table, for seed 0 only. (1) The window beats the best single checkpoint by +0.54 mAP [+0.29, +0.82] against e160 and +0.56 [+0.32, +0.77] against e180. (2) Fusion is order-dependent: added before min_keep and snap, hard + agn is +0.22; removed from FINAL it costs 0.06, with an interval that spans 0. (3) In the shift rows, snapping runs after the shift, so they include an interaction with rounding. (4) Last-stage and objectness removals keep the direction of the official steps, and λ2 has no measurable effect on F1 once weights are averaged (λ1 is +0.002 F1).

G. Dataset findings

G.1. Basic statistics

Table 66. Dataset statistics (recomputed from annotations unless marked).

Finding	Value
Images, split	9,392 labelled; local 7,512 / 940 / 940 by `prepare_course_data.py` seed 0, stratified by class, not grouped by scene or source. Official val 1,426, test 2,065
Boxes	32,533 total (train 26,015, val 3,323, test 3,195). Objects per image mean 3.46 / 3.54 / 3.40, median 2, max 41
Size	COCO small (original px area below 32^2) 69.7 %, medium 25.1 %, large 5.1 %. Side at 512 input, $\sqrt{wh}$: 18.4 % below 8, 50.5 % below 16, 62.4 % below 32, 93.2 % below 96; median 15.87
Class imbalance, images (local train)	Harbor 1,083; bridge 999; oil tank 987; ship 949; aircraft 946; playground 939; wind turbine 734; airport 520; car 355 (3.1:1)
Class imbalance, boxes (all)	Wind turbine 26.5 %, oil tank 17.2 %, car 12.6 %, bridge 12.2 %, harbor 8.1 %, playground 8.0 %, aircraft 7.8 %, ship 4.7 %, airport 2.8 %
Image resolution	Local train 5,578 at 512, 1,873 at 256, 48 at 224, a few non-square. Official val 1,067 at 512 (74.8 %), 349 at 256 (24.5 %), 6 at 224, one 500×302. Test: 512 bucket 1,621, 256 bucket 444. Ship almost all 256; aircraft mostly 512 with 224. $P(\text{class} \mid 224) = 1$ for aircraft; $P(\text{ship} \mid 256) \approx .496$; airport, car, playground, wind only at 512
Class median side at 512 (p10–p90)	Wind turbine 8.3 (5.5–15.0), bridge 9.9 (7.0–25.7), oil tank 11.5 (7.0–60.9), playground 14.8 (10.7–19.5), car 41.4, harbor 42.9, ship 55.1, aircraft 76.4, airport 146.0
Share below 16 px, share aspect above 3	Bridge 82.8 % / 3.3 %; playground 64.0 % / 0.1 %; wind 94.2 % / 0 %; oil 57.7 %; ship 0.5 % / 13.2 % (`data_audit.json`)
Object class = image class	Every annotated image has exactly one class in all three local splits (0 multi-class, 0 empty)
Boxes per training image, median / P(N≥2) / P(N≥5) / max	Aircraft 1 / .49 / .07 / 21; airport 1 / .33 / .01 / 5; bridge 2 / .59 / .24 / 23; car 8 / .95 / .80 / 28; harbor 1 / .47 / .07 / 15; oil tank 5 / .89 / .51 / 8; playground 1 / .43 / .09 / 19; ship 1 / .17 / .01 / 8; wind 8 / .89 / .70 / 41 (session tool output)
Source datasets	SARDet-100K, RSAR, M4-SAR, HRSID via SARFact, per the data package's `SOURCES_AND_LICENSES.md`

Table 67. Class composition on all 9,392 images (session tool output). Classes on official val and test are *predicted*, not labelled. Median $\sqrt{\text{area}}$ here is computed in a different scope from the side table above, so aircraft, harbor, oil tank and ship differ.

Class	Train images by size	Median $\sqrt{\text{area}}$	Train share	Val pred. share	Test pred.
Aircraft	1,182: 512 1,119 / 224 59 / 256 4	74.4	12.6 %	10.4 %	13.1 %
Airport	650: 512	146.0	6.9 %	13.1 %	9.4 %
Bridge	1,249: 512 1,011 / 256 238	9.8	13.3 %	8.4 %	10.2 %
Car	443: 512	41.4	not rec.	not rec.	7.4 %
Harbor	1,355: 512 982 / 256 373	41.1	14.4 %	8.3 %	6.2 %
Oil tank	1,233: 512 684 / 256 549	10.7	13.1 %	10.7 %	9.8 %
Playground	1,175: 512	14.8	12.5 %	13.2 %	12.5 %
Ship	1,187: 256 1,157 / 512 13	28.2	12.6 %	11.2 %	12.3 %
Wind turbine	918: 512	8.3	9.8 %	18.2 %	19.1 %

G.2. Annotation offset

Measurement. Moving every predicted box by −0.4 px at the 512 input raised mAP on 3 models and 2 holdouts by +0.4 to +0.7†; a C36 re-score confirms +0.48 / +0.62. Split by original resolution (C36, v3, recomputed):

Table 68. Shift sweep by original resolution, C36.

Shift at 512 input	0	−0.2	−0.4	−0.6	−0.8
Val, 512-px originals (712 images)	29.45	29.86	**29.94**	29.91	29.54
Val, 256-px originals (220)	43.35	43.44	43.57	43.70	**43.74**
Test, 512-px originals (705)	33.80	34.25	**34.38**	34.15	33.64
Test, 256-px originals (228)	40.51	40.79	40.96	40.97	**40.98**

A uniform shift at 512 input is −0.4 original px for 512 images but −0.2 original px for 256 images. The 256 curve keeps rising to −0.6/−0.8 (−0.3 to −0.4 original px), so the data fit a constant offset in *original* pixels of about −0.3 to −0.4, though the 256 curve is shallow. On A2 (corrected in training), 512 originals peak at −0.2 (val 30.81, test 34.88; −0.4 gives 30.65, 34.59) and 256 originals peak at −0.4/−0.6 (val 45.06, test 42.31). The residual is about −0.2 original px on both, so `--flip-bias -0.3` removes part of the offset, not all.

Cause. "Most likely the annotation coordinate convention" is inference: no annotation was inspected for it in that period. The integer lattice was found later (§E.8), and the link between the two was not established.

G.3. Annotation lattice

See Table 50 and §E.8. The lattice was measured on training labels only.

G.4. Near-duplicates

Nearest-neighbour cosine on frozen DINOv3 embeddings (session tool output; embeddings in `.runtime/emb/`).

Table 69. Near-duplicate bands.

Similarity	Official val → train	Train leave-one-out	Local val share / mAP	Local test share / mAP
≥ .99	9.9 %	9.9 %	9.5 % / 83.0	8.7 % / 83.7
.97–.99	29.1 %	27.8 %	24.4 % / 62.0	23.5 % / 60.1
.94–.97	33.0 %	32.7 %	33.5 % / 52.0	35.0 % / 55.8
< .94	28.0 %	29.6 %	32.7 % / 40.6	32.8 % / 39.9

Nearest-neighbour median: train leave-one-out .960, official val .961, local val .957, local test .955, vallike .962. Share above .98: .240, .257, .202, .209, .264. Playground: AP 86.9 / 80.8 on copies against 12.3 / 1.1 with no partner. Bridge images rarely have partners: 80 and 83 of 307–308 are "none", with 0 and 1 "copy"; AP stays about 4 to 9, except one test copy (86.5, $n = 1$). The interpretation given† is that the score is strongly driven by whether a near-duplicate exists. Near-duplicates were named as a likely reason for local above official as early as 10-20, but were measured only on 11-01.

G.5. Weak classes

Bridge and playground over the project: E00 AP 0.2–0.4 and 0.4–0.6; C48 v3 2.5 and 4.5; AA 1.7 and 3.7; V2 raw e120 4.2 and 12.8; SC1 2.9 and 5.5; lr1x (val) 4.78 and 22.94; W3 agn 4.2 / 6.2 and 22.3 / 18.9; TWIN1 FINAL (vallike) 8.1 and 33.0. Scopes differ between these values. Size alone does not explain weakness: wind is smaller and far better (lr1x AP 72.98). Bridge–playground confusion is 24 of 48 image-classification errors in lr1x; in lr05x bridge→playground 15, playground→bridge 11. A visual check found labels not misaligned, with bridges as 6–10 px dots at river and road crossings and playgrounds about 15 px in dense city†. Median best IoU of GT against predictions was 0†. Recall at IoU 0.1 was only 50–60 % on C48†. On E00, AP50 vs AP75 was aircraft 71 vs 16 and wind turbine 46 vs 4†; overall val 48.63 vs 18.32.

Table 70. Hypotheses about the data, not verified.

Claim	Status
Bridge and playground come mainly from M4-SAR, Sentinel-1, with labels drawn on optical images	Session notes†, external paper
M4-SAR SAR-only YOLOv8: AP50 bridge 44.9, playground 67.9, vs ours about 5 and 10	Session notes†; different data composition and splits
Other-class objects in a scene were left unlabelled (the basis for `--partial-neg`)	Gain was small (BP4, +0.11 local Δm without fusion). Later withdrawn as a confirmed main cause
Same-class objects are also unlabelled: many river crossings with 1–2 labelled, boxes where no river is visible, dense-urban playground blocks	From 12 rendered local val images†. No saved images, no count
"Natural ceiling" for these classes	Stated, withdrawn, and restated in a backup note. Evidence is the 12 images above; meanwhile playground reaches 22.94 at lr1x from 3.72 on AA by training longer and larger, with different scope

G.6. Local versus official

Size mix is similar between local and official val; class mix is not (Table 67). V2 e100: official-reference value on local test +44.81 against official 38.23, a mAP gap of 3.29 and an F1 gap of 0.53. Transfer for later models:

Table 71. Local to official transfer, mAP points.

Model	Local val	Local test	Official	Official − val
SC1 (shift −0.2)	41.21	43.17	40.85	−0.36
lr1x e130	52.27	n/r	49.66	−2.61
cos160 e150	52.63	53.23	50.32	−2.31
Context e150	53.60	53.05	50.19	−3.41

(continued)

Model	Local val	Local test	Official	Official − val
Context e160	53.69	n/r	50.15	−3.54

Macro-F1, local val / local test → official: cos160 e150 95.51 / 94.57 → 94.48; context e150 95.69 / 95.06 → 94.89.

H. Noise, seeds and bootstrap

H.1. C36 repeats

37 runs (seeds 11–46 plus seed 0), all with post-processing v3, sample std. SEED_47 (killed at e16) and SEED_48 (killed at start) have no metrics and are excluded.

Table 72. C36 repeats.

	36 seeds	37 runs (with C36)
Val F1	95.387 ± 0.295	95.391 ± 0.292
Val mAP	37.462 ± 0.354	37.456 ± 0.351
Test F1	95.212 ± 0.367	95.210 ± 0.362
Test mAP	38.903 ± 0.367	38.907 ± 0.363
L (mean)	35.558 ± 0.715	**35.556 ± 0.705**
L range	33.89 (seed 45) to 36.83 (seed 13)	
Raw L	31.723 ± 0.713	31.711 ± 0.707

Val mAP range 36.53–38.16, test 38.26–39.61. Per-class AP std (val + test mean, native post-processing; derived): bridge 0.20, playground 0.34, aircraft 0.81, airport 1.60, car 0.50, harbor 1.04, oil 0.67, ship 0.73, wind 0.84. These apply to C36 under native post-processing; comparisons in §D.2 use AA with A4 + A2 under final post-processing.

These seeds were not planned. The team had asked to prioritise more ideas over multi-seed review, and `watchdog.sh` queued fallback seeds automatically once the backlog emptied (first 10-22 00:57, last 10-23 15:42). Mean 2.05 h per seed, 73.72 h in total. The session later called this a problem for GPU use, though the numbers are useful.

H.2. What is single-seed

Every screen in Parts A and B is one seed, including A4, A2, AA, S5, N1, SC1, SC2, BP7 and F2. In 10-27 to 10-31 no experiment was repeated with a second seed. The one attempt (SC1 seed 1) was stopped by the team after minutes in favour of full-data training, and BP7 seed repeats were proposed and never started. The full-data 56.49 run has one seed, and the twins have one seed each, until the runs of §F finish.

Table 73. Other noise evidence.

Evidence	Value
Official reference	Δm 0.0 ± 0.9 (3 seeds), val (`docs/DETAILS.md`)
Terms: one reference-config run varies	About ±1 Δm; gains below about 2 need mean ± std over seeds
Rayner, 2 seeds[‡]	LoRA r16 +8.7/+8.1, r256 +10.0/+10.3, ViT-S +9.4/+9.2 (test, Ry)
Sampling SD of mAP	n=1,426: 0.765 (F1 0.543). n=2,065: 0.659 (F1 0.419) (session tool output)
SD of (test − val) from sampling	mAP 1.010, F1 0.686, so test-scale Δm SD 2.02 (session tool output)
Same trajectory, adjacent checkpoints (official Δm)	SC2 e130/e150 51.86/51.81; BP7 e110/e130 51.60/51.76; cos160 e130/e150 47.12/46.92; context e130/e150/e160 46.93/46.89/46.80; cos200 e160→e180 +0.0237, e180→e200 −0.12
Single-epoch swings, same seed, training eval	TWIN2 − TWIN1 between −2.00 and +2.63 mAP
Same weights, DDP resume	Pre-resume independent runs diverge max 0.000894 / 0.001289 in parameters (`$EXP/bp7_lr130/README.md`)
Early epochs vs final	BP7−SC2 +2.19, +2.57, then −0.10; hybrid and lr05x lead at e30, lose by e100
Holdout to holdout	Same paired weights: +2.26 (val), −0.12 (test) local Δm; harbor +3.66 vs −1.33 AP
Inference repeat	F2 re-inference: same classification, same 79,969 boxes, scores within 0.0024[†]. SC2 from low-precision raw: box 0.1 px, score 0.003[†]
Official vs local post-processing ratio	0.3 to 1.9, median about 1 (n=5, excluding RPN)[†]

Paired comparison on one image set	Differences of 0.1 to 0.3 Δm "real on val" for post-processing, but below about 0.5 to 1 Δm not for training runs[†]

H.3. Bootstrap method and its limits

Image-level resampling, paired across variants on the same draws. 500 resamples on the first pass, 300 for `boot2`, `boot3`, `boot5`, 400 for TWIN1, and 200 for the seed-0 ablation (Table 65). The evaluator was validated against pycocotools to 4 decimals. These intervals cover image sampling on the holdout at hand. They do not cover training seed or the difference between the holdout and the official sets, as the context result shows.

Over 30 variants were compared on the same holdouts, with no multiple-comparison correction. Both local holdouts had already been used for many decisions, so they are not a fresh blind test. A joint value for agn of about $0.004 \times 0.008 \approx 3 \times 10^{-5}$ across two "independent" holdouts was noted at the time[†]; it ignores this re-use. Many Phase 1 choices were also made on the val board itself.

The early yardstick of ± 0.3 mAP came from one pair (E1 last iterate vs C36); the measured single-run std is about 0.35 mAP, with a range of 1.6 points on val. The early validation rule was both holdouts up and mean mAP at least +0.4.

I. Rule compliance decisions

Terms as quoted in sessions and the official docs: at most 130M total and 40M trained parameters ("counting every weight it uses at test time, frozen or trained"; heads count; train-then-freeze does not escape it); "One model: no ensembles, no test-time augmentation", no tiling; one input per image at most 512×512; "Do not use other SAR datasets, or detection or classification models trained on them"; generic pretraining allowed and reported; "Do not label val or test images by hand and do not train on them in any way"; "Do not share code, data or predictions outside your own team". Phase 1: 5 per day, 2 reset cards, closes 11-01 23:59 SGT. Phase 2: 2 in total, 11-02 00:01 to 11-08 23:59.

Table 74. Rule decisions.

Item	Decision and reason	Evidence
Parameter budget	Hard guard in `train.py`, buffers included. Final 125,076,960 total, 39,867,360 trainable; F2 109,014,752 / 23,805,152	Code
Parameter averaging	Parameter averaging of one trajectory is allowed; it is distinct from averaging predictions	Team report on 11-01; `$EXP/autoresearch_20261031/README.md`
One forward pass	Stage-box mean, soft-NMS, voting, RPN 2000 re-use one forward pass; Cascade already averages class probabilities. TTA was looked at on 10-30 and not pursued. S8 (overlapping patches) judged legal: no upscale or tiling, same parameter count, compute reported	Session notes[†]; READMEs
256 px → 512 stretch	Kept. Same as reference code; the rule forbids exceeding 512	Session notes[†]
Pretrained models	No SAR-trained or other SAR data. DINOv3 LVD-1689M, IN21k ViT, ConvNeXt-T DINOv3 listed; LingBot judged generic pretraining (Base distilled from generic Giant), must be reported. DINOSAR excluded. Rayner's TerraMind runs are outside the entries; the Terms on the new site were read as allowing remote-sensing foundation models[†]	READMEs
External SAR data	On 11-01 the team asked whether external data could calibrate the local split. Refused by the assistant, citing the Terms and possible overlap with val/test; the team accepted vallike instead. Metadata of public SAR datasets was listed; nothing downloaded, per session notes[†]	Session notes
kNN label transfer	Local test only. Team declined to use it	§E.12
Official val and test inputs	Never trained on or labelled; normalisation from train only. Predicted class and size of official val (and test) set the vallike mix and reweight the gain estimate. No labels used. Selection on the val leaderboard is declared	`proxy_eval.py`, `phase2_gain.py`
Distillation	Avoided by team decision: must use a frozen generic pretrained model at most 130M	Session notes[†]
Mean ± std for gains below about 2 Δm	Terms. In Phase 1 only C36 satisfied it. §F addresses this for the final recipe	`$BK44/REPORT.md` §0.3
Uploads	All manual; assistants never submitted. Format check against the Evaluation page with `tools/check_submission.py`	`leaderboard_score.json` files
AI tools	Claude Code and Codex use is disclosed in the main paper	Main paper

J. Compute accounting

Hardware: 2 × NVIDIA RTX 6000 Ada, `nvidia-smi` total 46,068 MiB each (44.36–44.39 GiB per PyTorch), shared with other users. `train_hours` is wall time of the epoch loop including periodic evaluation; it excludes final evaluation and post-processing.

Table 75. GPU-hour total. All values are training only.

Block	GPU-h	Composition
Junkai line to 10-23	156.26	28 single-GPU runs 55.21 h + 4 DDP runs 13.67 wall h (27.33 GPU-h) + 36 seeds 73.72 h; 68 runs (derived from `train_hours`)
To 10-28	about 180	As stated in `$BK51/REPORT.md` §2. The runs listed for that span (SC1 2.64, SC2 attempts about 2.7, SC2_full150 12.76) reach about 174, so about 6 GPU-h of this figure are not itemised
10-28 to 11-01 about 21:00	about 159.4	`$BK56/run/gpu_hours_runs_since_2026-10-28.json` sums to 178.97, but it **double-counts** the cos200 resumes: the `_v2` row carries cumulative 14.14 h × 2 = 28.28, and the earlier stages (17.4 + 0.86 + 1.26 + small validation rows) repeat 19.62. Removing 19.62 gives 159.4
Subtotal to 11-01	**about 340**	Matches the backup report
TWIN1 + TWIN2 + AUG1 + DP1	about 48.3	28.4 + 18.8 + about 1.1 (derived)
Total at the 11-02 stop	**about 388**	About 208 since 10-28

Outside the total:

- All `RPT_*` runs of §F, which are still training. The total for the final paper is TBD.
- Rayner's and other teammates' runs on RTX 3080 (for example 24 ep 0.9–2.2 h each, 100 ep 3.9–8.7 h)[‡].
- `FINAL_v1` about 0.67 h, `FINAL_v2` 0.15 h, `V2L`/`V2F` 2 min each, SEED_47 about 0.95 h, SEED_48 about 0.2 h, A00/A01 about 9 min, and DDP hang debugging.
- All inference, tuning inference (about 55 s per split per variant under load), evaluation, smoke tests and CPU work. TWIN1 evaluation took about 14 min on one GPU.

A separate bottom-up sum for 10-27 to cos200 e130 gives about 173 GPU-h, training only. For the span of the hours file (excluding SC-era runs, extended to e200) the same method gives about 163, against 159.4 from the file; the file's wall-clock figures start at the first checkpoint, so they undercount (BP7 cos160: 11.90 vs 12.87 in `checkpoint_validation.json`).

Table 76. Per-run compute. Peak = `torch.cuda.max_memory_allocated` in GiB per GPU; excludes about 2.5 GiB process overhead.

Run	GPUs	Epochs, images	Train h	Peak	Note
E00	1	24, 7,512	1.02	7.44	Anomaly, §K
E01, E02, E04r	1	24	1.06, 1.13, 1.00	41.35, 42.05, 42.61	
E07	1	24	0.89	40.04	
D1, D2, D6, W3, D4	1	24	1.28, 1.12, 1.05, 1.19, 1.05	42.6, 42.6, 42.7, 39.5, 42.6	
L36	1	36	1.50	42.6	
C36, C48	1	36, 48	2.10, 2.71	42.6	
Screens	1	36	1.99–2.50	39.3–42.7	R2 2.50, S5 2.33
SEED_*	1	36	2.01–2.10	42.61	Mean 2.05
L1, L2	1	100, 8,452	6.15, 6.30	42.6	
V2	2	120, 8,452	3.86	37.95	37.2 img/s per GPU
S8	2	36, 7,512	2.68	40.39	14.2 img/s per GPU
F1, F2	2	100, 9,392	3.54, 3.59	37.95	36.4 img/s per GPU on F2
AA (no grad-ckpt)	1	36, 7,512	2.105	42.61	
SC1	1	36, 7,512	2.644	12.26	
SC2 aborted e8, attempt-1 e1, single-GPU e22, SC1s1	1		about 0.70 + 0.08 + 1.87 + 0.03		Derived from file times

(continued)

Run	GPUs	Epochs, images	Train h	Peak	Note
SC2_full150	2	150, 9,392	6.380	10.63	12.76 GPU-h
BP1, 2, 3, 4, 5	1	36, 7,512	2.887, 2.398, 2.401, 2.475, 2.888	12.65, 12.17, 12.17, 12.17, 12.66	13.05 GPU-h
BP7_spatial64, BP8, BP9	1	36, 7,512	2.649, 3.161, 2.926	14.45, 13.33, 12.65	8.74 GPU-h
BP7 full cos130	2	130, 9,392	6.130	11.77	12.26 GPU-h
lr1x, lr05x	2	130, 7,512	5.787, 6.031	11.77	
hybrid130	2	130, 7,512	9.315	11.77	Shared GPU
lr125x, partial e69	2		about 2.72 wall		About 5.4 GPU-h
cos160 local, partial e154	2		about 6.23 wall		About 12.5 GPU-h
Context 160	2	160, 7,512	7.930	11.77	15.86 GPU-h
BP7 full cos160 to e130	2	130, 9,392	6.436	n/r	12.87 GPU-h
LingBot to e64	2	64, 9,392	about 5.87 wall	n/r	About 11.7 GPU-h; shared with cos200
cos200 to e180	2	180, 9,392	**12.97**	12.07	25.9 GPU-h; the entry's last source checkpoint
cos200 to e200	2	200, 9,392	14.14	12.07	28.28 GPU-h
TWIN1 to e190	2	190, 8,400	14.21 wall		28.4 GPU-h; about 3.1 min/epoch, about 7 under contention[†]
TWIN2 to e145	2	145, 8,400	9.41 wall		18.8 GPU-h
AUG1 + DP1 attempts	2		about 32 min wall		About 1.1 GPU-h

Memory. The cos200 probe at e131 shows allocated 12.95×10^9 B = 12.07 GiB and reserved 13.15×10^9 B = 12.24 GiB. Inference peak allocated is 1.23×10^9 B at rpn 2000 (reserved 1.45 to 1.63×10^9 B) and 0.91×10^9 B at rpn 1000 (reserved 1.01×10^9 B).

Table 77. Inference time.

Setting	ms/img	Note
F1, F2	17.1, 17.7	One GPU, bf16, including post-processing; average of the batched loop, not single-image latency
SC2	16.1	`SC2_full150/stdout.log`
SC1 while training on the same GPU	60.5	Session notes[†]
SC2 RPN 2000, shared GPU	53.2	0.96 GiB
cos200 e180 rpn 2000, val	21.9	
e160 replay	21.4	
avg 160–180 / avg 100–160	23.4 / 24.5	
e200	25.4	
avg 100–200 x11, GPU shared	46.4	
avg 100–180 x9, val, GPU shared, batch 2	49.4	
e180 first export	50.8	
Test, final weights, `--batch-size 2`, GPU shared, memory cap 0.15	**60.1**	`$PH2/watch.log`
Batch 16, shared GPU, re-measured on the final weights	23	`$BK56` README: 23.2

Snap + check took 13 s for 89,110 boxes; min_keep rebuild + snap + check 52 s; averaging 30 s on CPU. Contention: throughput was 7.4 img/s per rank against a typical 28 while other users' workloads shared the GPU on 10-30, and about 13 to about 7 img/s per GPU on 11-02 for several hours[†]. Sharing with our own LingBot gave 11–12 img/s per rank, against 25–26 after it paused.

Infrastructure incidents not listed in Table 42. SC2 attempt 1 silently died as a child of its session, with about 3 h lost; fixed with `setsid nohup`. BP2 and BP3 failed at launch after a repository branch switch, because the queue copied the working tree; fixed by copying from the BP1 snapshot. A v1 resume check halted on a maximum parameter difference of

0.00138, because independent GPU runs differ before resume; v2 checks state equality instead. NCCL P2P hangs on this host: `NCCL_P2P_DISABLE=1` was needed for DDP launches, first on 10-20 and again from 11-01 21:36 on.

K. Corrections

Places where earlier internal notes were wrong, or where a claim is weaker than it first looked. "Backup report" means `$BK56/REPORT.md` and its companions, written on the night of 11-01, unless another backup is named. Those files are left as they were; this section takes precedence over them.

Table 78. Corrections to earlier notes.

#	Earlier statement	What the artifacts show
Scores, ranks and attribution of the final entry		
1	Final Phase 1 rank "2 of 8" (backup report, README, report page, project memory), from the 21:20 snapshot	Board at 22:16: Team 6 posted 65.85 at 21:25; 9 teams. Locked board: Team 7 **rank 3 of 9**, behind Team 9 (67.16) and Team 6 (65.85)
2	"Official val agreed: e100–180 beat e100–160 by 0.5 mAP"	Those rows differ in window *and* hard fusion. **The e100–180 window was never scored alone**
3	Window "+0.6 to +1.0", fusion "+0.5 to +0.7", stated as effects on val	**These ranges are inferred.** Only the sum of the last steps is measured: +1.58 from 54.83 to 56.41, then +0.08
4	min_keep "only appends boxes below every thresholded box, so AP cannot drop" (`final_after_watch.sh`, session note)	**Not guaranteed.** The code selects `topk` over all class scores and sets `score_thresh=0` before soft-NMS for that image. A check printed: every 56.49 box present in the mk100 rebuild: False, missing: 48. The measured gain is positive; the guarantee is not established
5	TWIN1 "average beats best single by about 0.6, paired bootstrap P = 1.00"	At the time `verdict.txt` compared only against e130. A bootstrap computed for this report gives +0.539 [+0.289, +0.822] against e160 (Table 65), on one seed
6	Heading "Tried on the official val" in the backup report	σ 0.35–0.6, shift other than -0.2, LR $\pm$25–50 %, class-loss weights and oversampling were local only or never submitted. "Box voting ($\pm$0.03)": local deltas are -0.006 to -0.029 mAP on SC1, and the official feedback has no number. "RPN 2000/3000 on old models (0)": AA local only; SC2 rpn2000 official unrecorded. "Dropping the 100-box cap (0)": SC1 local only
7	Backup report §7: AUG1 and DP1 running as challengers	Stopped at 21:48 on 11-01 and replaced by twins, all halted
8	λ2 kept as a gain	Official +0.04 Δm only; W3 local F1 is lower at λ2 on val; TWIN1 shows no effect after averaging
9	Team 6 strict band 58.3 (session note)	Recomputed from exact board values, $(10\,\text{mAP} - \text{AP50})/9$ gives 58.22. Ours 52.91 and Team 9 58.76 agree with the notes
Column labels and post-processing versions		
10	**L36 column label:** an older table's header says v3 and lists L36 at -2.79 against the C36 mean	L36's `metrics_pp.json` is v1. v3-equivalent: val 37.10, test 38.20, L 34.30, so **-1.26 against the C36 mean, not -2.79**. Under v1 for both: 32.77 vs 33.16 (-0.39)
11	**V2 column label:** an older table labels V2's curve v3	46.92 … 48.91 … 48.54 are **v2**. v3: 47.04 / 47.97 / 48.44 / 49.07 / 48.77 / 48.70. L1 and L2 are true v3, so they were compared against the wrong column. Conclusion unchanged, gap larger
12	Stage-box mean gain "+0.3 to +0.5"	`pp_eval.py` and session notes give +0.53 / +0.54 on C48. Range understated at the top
13	D6 "only +0.6 over E04r" in a table footnoted "without post-processing"	Raw +0.87; +0.63 is v1
14	"C36 about +36" in the v1 context	Raw 31.28, v1 33.16. The later v3 value is 35.46
15	C36 raw baseline in notes: val F1 94.92, mAP 35.30; test 37.24	`metrics.json`: val 95.01, 35.34; test 37.25. The likely source is a later CPU re-inference; not confirmed. Raw comparisons of A4, A2, B1, S8 at decision time used the note values
Screens and z-scores		
16	"A4, S5, A2 and N1 sit at 2–2.5 std"	A2 is 1.75 std with no shift. With -0.2 shift it is 2.34, but other rows are not given their best shift
17	A4 gain stated as +1.6, +1.5, +1.78, +1.69. A2: +1.32, +1.74, +1.23. A4b: +1.14, +1.04	Same cause: the baseline. +1.5 and +1.6 are rough; +1.78 and +1.32 are vs C36 seed 0; +1.69 and +1.23 are vs the mean; +1.74 uses A2's boosted shift
18	F2 parts confirmed before F2	The A4 and A2 single screens preceded F2, but the stacking check AA started after F2 ended. F2 went ahead without evidence that they stack. S5, N1, R1, S6 came too late to be considered
19	E09b's fall due to "drop path, LoRA+, stronger regularisation"	Not isolated. E09c never ran; D6 clears r64 only; α/r also differed (1 vs 2). E04r proves only that the combination was at fault
20	D3's fall due to the multi-level neck	No neck-only run exists. S5's positive 36-epoch result is the only support, at a different epoch count
21	S8 worse because of oil tank (-6.8); compute about 4×	Harbor -4.6 also fell. Measured compute 2.6×
22	"LoRA on MLPs changes nothing" (Rayner line, YOLO, r16, 24 ep)	Large gain on the Junkai line at r32 + norms + LR 3e-4. Different head and combination; never cross-tested

(continued)

#	Earlier statement	What the artifacts show
23	Reference would be averaged after R01	R01 never ran. L rests on a single-seed reference, unlike the official 3-seed reference
24	"F2 with no shift" judged worse than −0.2	Untested officially; the choice comes from A2 on local holdouts only
Part B claims		
25	Spatial prior gain	Three figures: +1.76 local (S5 vs C36 mean), +3.23 local (BP7_spatial64 vs AA), official −0.10 at e130 vs +2.19 / +2.57 at e40 / e80. Never tested alone on the board under a matched schedule
26	52.78 credited to the slower cosine	+1.02 also changed `--reproducible-loader`. Paired local evidence exists only at 7,512 images: official +1.29 at e130, raw mAP +0.56, F1 −0.32; one seed
27	BP1 (shared mode, small config) shows "no gain at equal epochs on the board"	The board test was classification mode on r80 / 1536 + SP64 at cos160, a different variant
28	LingBot 39.23 at e60 compared with BP7 48.40 at e80	Different epochs and schedules. Bracket: e40 32.89, e80 48.40. F1 94.02 is below all BP7 previews. The one same-schedule preview (`aaa.zip`, e50) has no score
29	Bridge and playground "natural ceiling", later "mostly unlabelled-duplicate problems"	Asserted, withdrawn, restated. Evidence is 12 images with no count (Table 70)
30	Bridge 53 %, playground 67 % RPN recall. "More proposals do nothing"	53/67 are AA with no artifact; 48.74/66.30 are SC1. The second half of proposals still adds coverage; these are different measures
31	SC1 per-class: oil 45.4 → 54.9, ship 41.1 → 38.6, harbor +3.8, wind +4.2, airport +2.7 (session note)	`$BK28/REPORT.md`: oil +8.3, ship −1.2, harbor +4.7, wind +3.9, airport +3.7. The note looks like val only and the report is the val + test mean; the note states no scope
32	"BP8 adds no parameters"	Relative to BP1. Against AA it adds 49,344
33	"Local overstates official by about 2 mAP"	Compares local *test* with official. Official − val gaps range from −0.36 to −3.54 (Table 71)
34	Official Δm noise ±0.5, ±1, 0.5–1; README "roughly ±0.5" for a full run	None measured on the big model. The README value is an extrapolation
35	"Local trends hold on Codabench"	Rests on 3 official points with confounded changes. SC1 has L 43.74 and official 28.90 at 36 epochs
Records		
36	F2's platform: Codabench (early note, commit `2524cc7`)	Later documents and `official_scores_phase1.json` say the new leaderboard. The raw source is not preserved. V2 and F1 are Codabench rows in that file, which says "same scorer"
37	Score dates in `official_scores_phase1.json`: V2 10-21, F1 10-23	Session timestamps: V2 10-20 22:23, F1 10-21 04:54. The contemporaneous ones are used in §B
38	67 runs / 155 GPU-h; 36 seeds	68 runs / 156.26 counting E00. 36 seeds and 37 runs are both right, with test mAP 38.90 vs 38.91
39	Submissions on 11-01: "ten", "13"	Differences of scope: 10 rows from e180 $\lambda 2$ on, 13 BP7, 14 with LingBot
40	What was posted before 11-01	Not established. A screenshot reading "+1.09 relative to posted entry" on 10-31 implies a posted 45.83 at that time, and the board still showed 45.83 at 17:21 on 11-01
41	AUG1 hang caused by `--compile` (log filename)	`nccl_test` logs point to NCCL P2P, and attempt 2 in eager mode also hung. Compile was not separately isolated. Also, notes credit a Codex session with launching AUG1 and DP1; the raw calls show Claude Code did
42	Challenger threshold +0.5 mAP (session note, backup report)	`PLAN.md`: +0.3 on both holdouts. Never applied
43	"Wider, earlier windows keep improving"	Corrected within the session to "not proven"; only "averaging itself about +0.25 mAP" stood
44	Local protocol validated by r = .98, .85	Those are post-processing deltas ($n = 5$). The six-model check has lower r and reverses context vs plain (Table 56). No report of it to the team was found
45	GIoU + IoU branch "real strict-IoU lead"	Read at e80–e120 (+0.11 to +0.62). Later e130, e140 are +0.15, +0.04; training-default post-processing; unresolved
46	"Test may be easier": other teams' test mAP +0.3 to +1.3 over val	Sampling SD of test − val is 1.0 mAP, with $n = 2$. Within noise
47	Snap gain +0.21 / +0.36; +0.18 / +0.31; +0.03 / +0.05	Consistent after rounding: +0.177 + 0.027 = +0.204; +0.309 + 0.053 = +0.362
Compute		
48	`gpu_hours_runs_since_2026-10-28.json` sums to 178.97	Double-counts cos200. About 159.4 after removal (§J)
49	"48 GB", "12.1 GB", "38.0 GB"	46,068 MiB by `nvidia-smi`, 12.07 GiB, 37.95 GiB: the same quantities in mixed units
50	Peak E00 7.44 GiB vs E01 41.35 GiB on the same recipe with DINOv3; `bench/vit` 40.5 with IN21k; docs 11–17 GiB. S8 quoted at 16.7 and 43.6	Cause not found. S8's artifact value is 40.39 GiB; 43.6 is reserved memory from `nvidia-smi`
51	`--compile` "about 1.3× faster"	A01 measured +12 % (37.0 → 41.5–41.9) over 2 epochs; no longer measurement exists
52	Inference "23 ms", "0.8 GB at batch 1"	Logs range 21.4 to 60.1 ms and 0.91 to 1.23×10^9 B allocated, by setting. Commit `cbbe218` quotes soft-NMS at 37 ms/image on CPU and 337 on GPU in an earlier test; not reconciled

(continued)

#	Earlier statement	What the artifacts show
Team drafts		
53	An earlier team draft (10-29): "a 5-fold split cross-checks"; seed noise about ± 1	No 5-fold run exists; the jobs were cancelled. Measured on C36: ± 0.70

Recorded only in session notes, with no artifact found: the C48 corner, x/y and original-px grid; C48 recall at IoU 0.1; per-class AP50 vs AP75 on E00; per-size AP of E00, E01, E02, E04r, E07, E09b (`tools/size_ap.py` can regenerate them); the λ table on long models; soft-NMS flatness; the AA error-analysis table; the fusion table on AA; the 12 rendered images; M4-SAR AP50 values; numbers from public forks.

Could not be verified from this server: any Rayner-line or teammate number (no run folders here); the content of team screenshots; research-workflow reports whose tool output was removed; whether `V2_e70_val.zip` was submitted.

L. What was not established

- Which file or files were uploaded for Phase 2 (§E.13); the board shows the score, not the file.
- The official effect of lattice snapping and min_keep. Whether the lattice holds on val and test labels.
- The official effect of the e100–180 window without fusion. The board's split between window and fusion.
- Seed variance of the final recipe at full length, until the runs of §F finish.
- Whether GIoU + IoU branch, photometric augmentation, smaller anchors or drop path help on the full recipe. TWIN2 stopped at e145; TWIN3–5 never ran. AUG1 stopped at e3.
- The split of SC2's +7.02 between scaling and epochs, and between LoRA and head.
- The spatial prior's effect on the board under a matched schedule.
- Constant LR or a WSD tail, and LR $\times 1.25$ to completion.
- A pure schedule effect on full data, since cos130 $\rightarrow$ cos160 also changed the loader.
- Which part of E09b caused the fall. Copy-paste alone. The multi-level neck alone. The cause of R2's drop and of S8's oil tank and harbor drop.
- The cause of the annotation offset, and its link to the lattice.
- The contributions of class mix and near-duplicate share to the vallike level.
- Whether unlabelled same-class objects limit bridge and playground.
- How to close the strict-IoU gap of about 5 mAP to the leading teams.
- The final state of the Phase 2 board.

M. Reproduction

Environment. Conda env `cs701bench`, torch 2.9.0+cu128, timm 1.0.24; exact versions in `$BK56/run/pip_freeze.txt`. `prepare_course_data.py` converts the Hugging Face download into `dataset/SARFact-Course-20K`. Differences between the training snapshot and the backup code are in `$BK56/run/code_diff_training_snapshot_to_backup.diff`. Below, `$RUNS` and `$BK56` are as in §A, `$DATA` is the converted dataset, `$COURSE` is `cs701_data/cs701-sar-course-data`, and commands run from the repository root unless stated.

Final model (`$BK56/train_BP7_cos200.sh`; all 9,392 labelled images, 2 GPUs, stops cleanly after e180 with the 200-epoch schedule intact). The original went on to e200 and ran in stages with `--resume`, plus `tools/train_gpu_limited.py` to cap memory. Each resume restored model, AdamW, schedule and both ranks' RNG.

```
cd $BK56/code
NCCL_P2P_DISABLE=1 PYTORCH_ALLOC_CONF=expandable_segments:True TMPDIR=<dir on a large disk> \
torchrun --standalone --nproc_per_node 2 train.py \
  --data "$DATA" --out "$BK56/runs" --name BP7_cos200 \
  --train-split train+val+test --eval-splits '' --seed 0 --batch-size 16 --workers 1 \
  --epochs 200 --schedule-epochs 200 --warmup-epochs 1 --stop-after-epoch 180 \
  --save-every 10 --save-training-state --reproducible-loader \
  --backbone dinov3 --init pretrained --adapt lora \
  --lora-targets attn.qkv,attn.proj,mlp.fc1,mlp.fc2 --lora-rank 80 --lora-alpha 160 \
  --cascade-rep 1536 --spatial-prior 64 --train-norms \
```

```
  --lr 1e-4 --backbone-lr 3e-4 --weight-decay 0.05 \
  --cascade --aligned-roi --rpn-nwd 8 --anchor-offset 0.5 \
  "--anchor-sizes=6,8,11;16,19,23;32,38,45;64,76,91;128,152,181" \
  --cascade-size-iou --flip-bias -0.3 --grad-ckpt --eval-every 10 --eval-batch-size 4
```

The log should start with `parameters: 125,076,960 total, 39,867,360 trainable`. Training is not bit-deterministic under DDP on GPU, so a retrained trajectory differs from the original. The backup README's $\pm 0.5\,\Delta m$ for such a run is an extrapolation, not a measurement.

Averaging. Equal weights over 234 trained tensors, FP64 accumulation, FP32 output, about 30 s on CPU. `tools/average_ckpts.py` has the original checkpoint paths of the four directories built in; `$BK56/average_checkpoints.py` takes `--dir` for a retrained run.

```
python tools/average_ckpts.py --epochs 100 110 120 130 140 150 160 170 180 \
  --out .runtime/avg/avg_e100-180.pt
# for a retrained run:
python $BK56/average_checkpoints.py --dir $BK56/runs/BP7_cos200 \
  --out $BK56/runs/BP7_cos200/avg_e100-180.pt
```

Phase 1 entry, 56.49 (`$BK56/make_submission.sh`; about 1 min on one GPU). Re-scoring `weights/val_raw.pt` on CPU gives a byte-identical `classification.csv` and `detection.json`. Predicting afresh at batch 16 gave identical classification and 59,668 boxes instead of 59,711, since bf16 batch effects move a few boxes across the threshold. Compare files inside the zips, because a zip records file times.

```
cd $BK56/code
python predict.py --run ../weights --weights model.pt --images $COURSE/val --batch-size 16 \
  --rpn-topn 2000 --obj-pow 0 --soft-sigma 0.5 --stage-box mean --cls-lambda 2 --shift -0.2 \
  --fusion agn --gamma 1 --topk 9 --out BP7_avgE100-180_val_agn.zip
python tools/check_submission.py BP7_avgE100-180_val_agn.zip $COURSE/val/images.json
```

FINAL zip for test (`$PH2/watch_test_release.sh`, `$PH2/final_after_watch.sh`). First a forward pass that caches raw outputs and writes the plain-recipe zip, then the min_keep rebuild, snap and format check. Run from the repository root.

```
RUN=$RUNS/BP7_full9392_cos200_e160to200_s0_20261101_v2/full_train_e200
TJ=$COURSE/test/images.json
python tools/train_gpu_limited.py --memory-fraction 0.15 predict.py --run $RUN \
  --weights .runtime/avg/avg_e100-180.pt --images $COURSE/test --images-json $TJ \
  --batch-size 2 --workers 2 --rpn-topn 2000 --obj-pow 0 --soft-sigma 0.5 --cls-lambda 2 \
  --shift -0.2 \
  --fusion agn --gamma 1.0 --topk 9 --raw-out .runtime/avg/avg_e100-180_test_raw.pt \
  --out submissions/PHASE2_shot1_avg100-180x9_rpn2000_obj0_soft05_cls2_agn_g1_k9.zip
python predict.py --run $RUN --from-raw .runtime/avg/avg_e100-180_test_raw.pt \
  --images $COURSE/test --images-json $TJ --rpn-topn 2000 --obj-pow 0 --soft-sigma 0.5 \
  --cls-lambda 2 --shift -0.2 --fusion agn --gamma 1.0 --topk 9 --min-keep 100 \
  --out submissions/PHASE2_shot1_mk100.zip
python tools/snap_submission.py submissions/PHASE2_shot1_mk100.zip $TJ \
  submissions/PHASE2_shot1_FINAL_avg100-180x9_agn_mk100_SNAP.zip
python tools/check_submission.py submissions/PHASE2_shot1_FINAL_avg100-180x9_agn_mk100_SNAP.zip $TJ
```

Expected: 2,065 images, 0 boxes outside, 155,989 boxes, of which 62,708 moved to integers and 11,812 wind boxes moved to a lattice. SHA256 prefix of the original: `c90b0d5b`.

Twins and report runs. TWIN1: `$PH2/launch_twin.sh <name>`, with extra flags for variants. Report runs: `python $RPT/queue.py` (`--smoke` for the 64-image check). Scoring: `$RPT/eval_tools/ablate_pp.py <rawdir> <out.json> [nboot]` and `$RPT/eval_tools/analyze_loo.py`. Tables: `python report_final/gen_report.py`, then `tectonic supplement.tex` from `report_final/`.